Natural boundaries of the full square-lattice Ising susceptibility in two complex coupling variables

Endpoint geometry and constrained transport

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Abstract

We determine natural boundaries for the bulk zero-field susceptibility of the infinite square-lattice Ising model with two independent complex couplings. An endpoint criterion identifies explicit high- and low-temperature components of the spectral complement in $\mathbb{C}^2$. The physical germs extend holomorphically to these domains, and every boundary point obstructs continuation. At selected regular points the estimates apply along holomorphic transverse curves with nonramified edge coordinate and are locally uniform in complex dispersion parameters. At negative edges of the odd sum, pole periods on the real cycle isolate a single particle index at selected Taylor orders. A constrained transport estimate treats both parities on the collinear isotropic slice. Exact pullbacks give the natural boundaries for every fixed positive ratio, and $s = \sinh z$ gives the isotropic disk and exterior germs.

1 Main results and the physical observables

The zero-field susceptibility sums spin correlations over the entire square lattice. Its form-factor expansion leads to a noncancellation problem for an infinite series of singular integrals [3, Section 3.2][4, Section 6]. With two independent couplings the analytic problem includes both the joint continuation and the continuations on the physical temperature lines.

The variables K_x, K_y are independent coupling coordinates. Put

$$z = 2K_x = x + \mathrm{i}y, \qquad w = 2K_y = u + \mathrm{i}v,$$
$$S = \sinh z, \quad T = \sinh w, \quad A = \cosh z \cosh w, \quad p = ST.$$

For real positive couplings, $\mathcal{X}_H$ is the sum of all two-spin correlations in the unique high-temperature Gibbs state; $\mathcal{X}_L$ is the sum of all connected two-spin correlations in the infinite-volume plus phase. Section 3 gives the physical definitions and identifies their form factors.

Define the spectral set by

$$\Sigma = \{(K_x, K_y) : A = Sa + Tb \text{ for some } a, b \in [-1, 1]\}.$$

The two domains used in this paper are explicitly

$$\mathcal{D}_H = \{|y| < \pi/2,\ |v| < \pi/2,\ \cosh x \tanh|u| < \cos y, \quad \cosh u \tanh|x| < \cos v\}, \tag{1.1}$$
$$\mathcal{D}_L = \{x > 0,\ u > 0, \quad \cosh x \tanh u > |\cos y|, \quad \cosh u \tanh x > |\cos v|\}. \tag{1.2}$$

In these coupling coordinates $\mathcal{D}_H$ is unbounded along the decoupled directions and its positive-ratio temperature slices are bounded.

Theorem 1.1 (Joint physical natural boundaries)**.** *The following assertions hold for both phases of the full bulk reduced susceptibility.*

1. *The sets $\mathcal{D}_H$ and $\mathcal{D}_L$ are precisely the components of $\mathbb{C}^2 \setminus \Sigma$ containing, respectively, the real positive regions $0 < p < 1$ and $p > 1$. They are contractible and pseudoconvex. On them $|p| < 1$ and $|p| > 1$, respectively.*

2. *The physical germs extend to single-valued jointly holomorphic functions $\mathcal{X}_H \in \mathcal{O}(\mathcal{D}_H)$ and $\mathcal{X}_L \in \mathcal{O}(\mathcal{D}_L)$. Their complete odd and connected-even form-factor sums converge normally on compact subsets, with every fixed mixed complex derivative.*

3. *At every point of the respective boundary, there is no holomorphic function on a neighborhood agreeing with that physical continuation on an accumulating connected component of its domain side. These are thus natural boundaries also for continuation in unramified Riemann domains over $\mathbb{C}^2$, reached from the given physical germ.*

4. *At a dense set on every regular boundary face, the complete physical sum is nonextendable along every holomorphic transverse curve on which the local edge parameter is nonramified. The estimates are locally uniform for a fixed selected phase and compact admissible subfamilies of bounded holomorphic complex-parameter families on common polydisks in a regular-edge chart. The collinear equal-coupling restriction has the dense local obstructions of Proposition 6.10.*

Here regular faces and their edge parameters are specified by Definition 2.5 and Proposition 2.7.

For every finite $\lambda > 0$, put $\iota_\lambda(\zeta) = (\zeta/2, \lambda\zeta/2)$. Let

$$P_\lambda(a) = \cosh a \tanh(\lambda a), \qquad Q_\lambda(a) = \cosh(\lambda a) \tanh a, \qquad Y_\lambda = \frac{\pi}{2\max(1,\lambda)}.$$

Both functions on $[0,\infty)$ have increasing inverses. Define

$$\begin{aligned} H_\lambda(b) &= \min\{P_\lambda^{-1}(\cos b), Q_\lambda^{-1}(\cos(\lambda b))\}, && |b| < Y_\lambda, \\ L_\lambda(b) &= \max\{P_\lambda^{-1}(|\cos b|), Q_\lambda^{-1}(|\cos(\lambda b)|)\}, && b \in \mathbb{R}, \\ \Omega_{H,\lambda} &= \{a + \mathrm{i}b : |b| < Y_\lambda,\ |a| < H_\lambda(b)\}, \\ \Omega_{L,\lambda} &= \{a + \mathrm{i}b : a > L_\lambda(b)\}. \end{aligned}$$

Theorem 1.2 (Exact recovery of every positive-ratio slice)**.** *For every finite positive λ,*

$$\iota_\lambda^{-1}(\mathcal{D}_H) = \Omega_{H,\lambda}, \qquad \iota_\lambda^{-1}(\mathcal{D}_L) = \Omega_{L,\lambda}.$$

The restrictions of the two joint physical functions are the respective physical reduced susceptibilities, holomorphic on these exact domains, and nonextendable through every boundary point in the componentwise sense of Theorem 1.1. They retain the complete lattice sum and every odd or nonzero even particle order. On a physical line with fixed $J_x > 0$, $J_y = \lambda J_x$, the same conclusions hold for the energy-field response

$$\chi(\zeta) = \frac{\zeta}{2J_x}\,\mathcal{X}(\iota_\lambda(\zeta)).$$

The proof of nonextension on a slice uses the transverse-curve estimates of Theorem 1.1 and the local isotropic estimates at equal coupling.

Corollary 1.3 (Exact isotropic descendants)**.** *Set $\lambda = 1$ and $s = \sinh\zeta$. The reduced HT physical germ is single valued and holomorphic on $|s| < 1$, with value one at zero; the reduced connected-plus-phase LT physical germ is single valued and holomorphic on $|s| > 1$. Every point of $|s| = 1$ is a natural-boundary point for each germ. Their complete normalized representations are*

$$\mathcal{X}_H(s) = \frac{(1-s^4)^{1/4}}{s} \sum_{N\geq 1,\ N \text{ odd}} C_N(1/s), \qquad \mathcal{X}_L(s) = (1-s^{-4})^{1/4} \sum_{N\geq 2,\ N \text{ even}} C_N(s),$$

where the coefficients are the full-site integrals in Section 6. The LT onsite term is $1-(1-s^{-4})^{1/4}$. The map sinh *is a biholomorphism from $\Omega_{H,1}$ to the disk and is the universal covering from $\Omega_{L,1}$ to the exterior, with deck translations $\zeta \mapsto \zeta + 2\pi\mathrm{i}k$. The energy-field response uses $\beta = \zeta/(2J)$ on the physical temperature continuation.*

1.1 Relation to earlier work

The unsquared anisotropic Nickel equation appears in [4, Section 6]. We use its spectral complement to identify the two physical domains. The real noncritical spin form factors and the Gibbs-state and ferromagnetic results used to identify the physical observable are given in Section 3.

The endpoint locus contains the spectral boundary and determines the two physical components. Double contours give the jointly holomorphic susceptibilities. A fixed Vandermonde subproduct controls the tail at regular resonant edges. At negative odd-particle edges, exact residues on the real angular plane, an angular concentration estimate and a complex beta period isolate a single index in the boundary Taylor coefficients. On the collinear slice a constrained transport based on an analytic variance controls the tails for both parities. The first resonance is evaluated on the constrained real cycle.

The decoupled-axis restriction $\mathcal{X}_H(K_x, 0) = e^{2K_x}$ is entire even though the joint domain has boundary points on that line. For each unequal positive ratio we obtain a dense set of nonramified transversals on the slice and apply the estimates with their holomorphic anisotropy B. Section 6 treats the full sum at equal coupling.

AI Declaration. ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

Contents

1 Main results and the physical observables **1**
1.1 Relation to earlier work . . . 3

2 The physical components from endpoint incidence **5**
2.1 The two entire physical components . . . 5
2.2 Regular joint faces and their density . . . 7
2.3 Every positive-ratio slice . . . 9
2.4 The local collar and propagation to exceptional points . . . 10
2.5 The isotropic disk and exterior covering . . . 11

3 The physical observable and joint holomorphy **12**
3.1 Real-temperature form factors and state identification . . . 12
3.2 Double contours and the full lattice . . . 13
3.3 Energy-field normalization . . . 15

4 Regular spectral edges and transverse curves **16**

5 The full odd sum at a regular negative spectral edge **21**
5.1 Statement, branch, and the physical side . . . 21
5.2 A uniform Gaussian numerator bound and all boundary jets . . . 22
5.3 An exact reduction to simple-pole periods . . . 24
5.4 Bounds for every winding and every particle index . . . 25
5.5 Uniform scalar poles on the original real plane . . . 26
5.6 The complete residue and its angular measure . . . 28
5.7 An exact complex beta period and the two first windings . . . 30
5.8 Integer selection of a single dominant particle index . . . 32
5.9 Exhausting the remaining indices and windings . . . 34
5.10 Nonextension and uniformity . . . 36

6 The isotropic restriction on the complete constrained cycle **36**
6.1 The dense arithmetic family . . . 37
6.2 One analytic tail estimate for both isotropic parities . . . 38
6.2.1 A deformation of the complete constrained torus . . . 38
6.2.2 The mixed branch populations . . . 39
6.2.3 The regular inverse and a cell Gaussian bound . . . 40
6.2.4 The exact current and its joint jets . . . 42
6.2.5 Coarea with coupled occupancies . . . 46
6.2.6 One sum for all higher particle orders . . . 47
6.3 The finite complement and the original prime family . . . 48
6.4 Dense local obstructions and the exact physical identities . . . 51

7 Proofs of the joint, slice and isotropic theorems **51**
7.1 The complete joint obstruction . . . 51
7.2 All physical ratios, including the collinear one . . . 52
7.3 Derivation of the complete isotropic theorems from the slice theorem . . . 53

2 The physical components from endpoint incidence

Put

$$z = 2K_x = x+iy, \quad w = 2K_y = u+iv, \quad S = \sinh z, \quad T = \sinh w, \quad A = \cosh z \cosh w, \quad p = ST,$$

and retain the unsquared spectral set

$$\Sigma = \{(z,w) : A = Sa + Tb \text{ for some } a, b \in [-1,1]\}. \tag{2.1}$$

The change between (z,w) and the coupling coordinates is invertible. Compactness of the representing square makes Σ closed. Define

$$F_1 = \cosh^2 u \cos^2 y - \cosh^2 x \sinh^2 u, \qquad F_2 = \cosh^2 x \cos^2 v - \cosh^2 u \sinh^2 x, \qquad E = \{F_1 F_2 = 0\}. \tag{2.2}$$

Only the boundary of the spectral set is needed to determine its two physical complementary components. The following endpoint criterion avoids a classification of the spectral interior.

Lemma 2.1 (Endpoint boundary criterion)**.** *The set E is exactly the set of parameters admitting a representation in* (2.1) *with a or b equal to 1 or -1. Consequently*

$$E \subset \Sigma, \qquad \partial\Sigma \subset E. \tag{2.3}$$

These statements hold for independent complex z, w, including zero real parts, zeros of S or T, and real-linear dependence of S, T.

Proof. If $S \neq 0$, $\tanh(z/2)$ is finite and nonzero. For $\epsilon \in \{1,-1\}$ put $r_\epsilon = e^{\epsilon w} \tanh(z/2)$. Direct multiplication gives

$$\frac{r_\epsilon + r_\epsilon^{-1}}{2} = \frac{A - \epsilon T}{S}, \qquad |\tanh(z/2)|^2 = \frac{\cosh x - \cos y}{\cosh x + \cos y}. \tag{2.4}$$

Both terms in the last ratio are positive when $S \neq 0$. Therefore

$$|r_\epsilon| = 1 \quad \Longleftrightarrow \quad \cos y = \epsilon \cosh x \tanh u.$$

The preimage of $[-1,1]$ under $(r + r^{-1})/2$ is the unit circle. Thus $A = Sa + \epsilon T$ with $a \in [-1,1]$ is equivalent to the last equation. Taking the union over ϵ gives exactly $F_1 = 0$. If $S = 0$, then $x = 0$, $y \in \pi\mathbb{Z}$, and $F_1 = 1$. Also $A^2 - T^2 = 1$, so no such endpoint representation exists. This proves the equivalence globally. Exchange z, w for F_2.

For $P \in \partial\Sigma$, consider the real-linear map $L_P(a,b) = S(P)a + T(P)b$. If it has rank two and the representing pair is in the open square, continuity of $L_K^{-1} A(K)$ would make P an interior point of Σ. Its pair must therefore have an endpoint coordinate. If the rank is at most one, the affine fiber of any representing pair has a nonzero kernel direction; following that direction inside the closed square reaches its boundary. There is again an endpoint representation. The equivalence just proved gives $P \in E$. □

2.1 The two entire physical components

For $a, b \geq 0$ write

$$r_1(a,b) = \cosh a \tanh b, \qquad r_2(a,b) = \cosh b \tanh a.$$

Both functions are nondecreasing in each argument and strictly increasing in their hyperbolic-tangent argument. Also

$$r_1(a,b)^2 - 1 = \frac{\sinh^2 a \sinh^2 b - 1}{\cosh^2 b}, \qquad r_2(a,b)^2 - 1 = \frac{\sinh^2 a \sinh^2 b - 1}{\cosh^2 a}. \tag{2.5}$$

Theorem 2.2 (Physical components and branches)**.** *The component of* $\mathbb{C}^2 \setminus \Sigma$ *containing the positive real region* $0 < p < 1$ *is*

$$\mathcal{D}_H = \{|y|, |v| < \pi/2, \quad r_1(|x|, |u|) < \cos y, \quad r_2(|x|, |u|) < \cos v\}. \tag{2.6}$$

The component containing the positive real region $p > 1$ *is*

$$\mathcal{D}_L = \{x, u > 0, \quad r_1(x, u) > |\cos y|, \quad r_2(x, u) > |\cos v|\}. \tag{2.7}$$

Both domains are contractible, and

$$|p| < 1 \text{ on } \mathcal{D}_H, \qquad |p| > 1 \text{ on } \mathcal{D}_L. \tag{2.8}$$

Proof. Simultaneous dilation of all four real coordinates toward zero preserves the strict inequalities of $\mathcal{D}_H$: their left sides decrease and their right sides increase. Hence this domain is star shaped about zero. Its closure is exactly

$$|y|, |v| \le \pi/2, \quad r_1(|x|, |u|) \le \cos y, \quad r_2(|x|, |u|) \le \cos v. \tag{2.9}$$

Indeed, every point satisfying these weak inequalities is moved into the strict domain by every dilation factor $t < 1$. For an active first inequality, either $u \neq 0$, in which case $\tanh(t|u|)$ strictly decreases, or $u = 0$ and $|y| = \pi/2$, in which case $\cos(ty) > 0$; the second inequality is the exchanged calculation. Thus every finite boundary point has $F_1 F_2 = 0$.

For $\mathcal{D}_L$, increasing both real parts strictly opens every weak inequality, so its closure is exactly

$$x, u \ge 0, \quad r_1(x, u) \ge |\cos y|, \quad r_2(x, u) \ge |\cos v|. \tag{2.10}$$

If both F_i are strictly negative at such a point, then $x, u > 0$ and the point is interior. Its finite boundary is therefore also in E. To contract the domain, fix $R > 0$ with $\sinh R > 1$. First increase (x, u) to $(\max(x, R), \max(u, R))$, then move to (R, R) within $[R, \infty)^2$, and finally contract (y, v) to $(0, 0)$. The first move preserves the inequalities by monotonicity; during the other two, both r_i are at least $\sinh R > 1$. These three continuous homotopies give a contraction.

On the respective domains both F_i have a strict common sign, so neither domain meets E. By (2.3), its intersection with Σ is relatively open and closed. Both domains are connected. Zero is outside Σ, and the positive real point (R, R) above is outside Σ, since

$$(A - S - T)(A + S + T) = (p - 1)^2 > 0$$

there and $A + S + T > 0$. Hence both domains are disjoint from Σ. Their boundaries lie in Σ, so each is an entire complementary component. Equation (2.5) identifies the indicated positive real regions.

Finally,

$$|p|^2 = (\cosh^2 x - \cos^2 y)(\cosh^2 u - \cos^2 v). \tag{2.11}$$

If $F_1, F_2 > 0$, the two factors are respectively less than $\cosh^2 x / \cosh^2 u$ and $\cosh^2 u / \cosh^2 x$; if both are negative, both inequalities reverse. This proves (2.8). □

The binomial series define $(1 - p^2)^{1/4}$ on $\mathcal{D}_H$ and $(1 - p^{-2})^{1/4}$ on $\mathcal{D}_L$, with their positive real physical values. The HT transfer zeros are exactly the complex axes $z = 0$ and $w = 0$; neither transfer variable vanishes on the LT domain. The corner identity

$$A^2 - (\epsilon S + \delta T)^2 = (p - \epsilon\delta)^2, \qquad \epsilon, \delta \in \{1, -1\}, \tag{2.12}$$

will also be useful.

Lemma 2.3 (Canonical decaying root)**.** *For $(z, w) \notin \Sigma$ and real q, the equation*

$$S\rho^2 - 2(A - T\cos q)\rho + S = 0 \tag{2.13}$$

has a unique root with $|\rho| < 1$, interpreted as zero at $S = 0$. It is holomorphic in both couplings, locally jointly holomorphic with the complex angle, and has a uniform modulus gap below one on each compact parameter set and the real angular circle. On either positive real physical region it is the positive decaying root.

Proof. For $S \neq 0$, $W = (A - T\cos q)/S$ is outside $[-1, 1]$ by the unsquared definition of Σ. The reciprocal roots of $\rho + \rho^{-1} = 2W$ are distinct and neither is on the unit circle; exactly one is inside. Its local implicit-function branches agree by uniqueness. At $S = 0$, $A - T\cos q \neq 0$, so the implicit-function theorem extends that root through zero. Compactness gives the gap and a common thin complex angular neighborhood. On positive real noncritical parameters, the corner identity gives $A - S - T > 0$, hence $W > 1$ and $0 < \rho < 1$. □

Proposition 2.4 (Pseudoconvexity)**.** *Every component of $\mathbb{C}^2 \setminus \Sigma$, in particular $\mathcal{D}_H$ and $\mathcal{D}_L$, has a continuous plurisubharmonic exhaustion.*

Proof. The continuous function

$$m(K) = \min_{a,b\in[-1,1]} |A - Sa - Tb|$$

is positive on a complementary component and zero exactly on Σ. There $-\log m$ is the continuous locally bounded supremum of the pluriharmonic functions $-\log|A - Sa - Tb|$, so it is plurisubharmonic by the submean inequality on complex lines. Thus

$$|K_x|^2 + |K_y|^2 + \max(0, -\log m(K))$$

is a continuous plurisubharmonic exhaustion: it diverges at infinity and at every finite boundary point. A bounded sublevel has m bounded away from zero, so all its limit points remain in the same component. □

2.2 Regular joint faces and their density

Definition 2.5. A boundary point is a regular joint edge when $S, T \neq 0$, $D = \Im(\overline{S}T) \neq 0$, exactly one of its representing coefficients is an endpoint, the other lies in $(-1, 1)$, and $p^2 \neq 1$.

For face regularity we use the following endpoint identity. Suppose $F_1 = 0$ and use its endpoint representation $A = Sa + \epsilon T$, $a \in [-1, 1]$. Set $K = \cosh z\, T - \epsilon \cosh w$. Then

$$K^2 = S^2(a^2 - 1), \qquad \cos y = \epsilon \cosh x \tanh u.$$

Writing $c = \cosh x$, $C = \cosh u$, $H = \sinh u$, we obtain

$$\Re(\overline{K}T) = c\cos y\,|T|^2 - \epsilon HC = \epsilon\frac{H}{C}\{c^2(C^2 - \cos^2 v) - C^2\} = -\epsilon\tanh u\, F_2.$$

Since $K/S = \pm i\sqrt{1 - a^2}$, including the zero case, this proves

$$D^2(1 - a^2) = \tanh^2 u\, F_2^2. \tag{2.14}$$

Also direct substitution in (2.11) gives

$$|p|^2 = 1 - \frac{F_2}{\cosh^2 u} \quad (F_1 = 0), \qquad |p|^2 = 1 - \frac{F_1}{\cosh^2 x} \quad (F_2 = 0). \tag{2.15}$$

Thus $x, u \neq 0$ and exactly one vanishing F_i imply all the regular-edge conditions on either physical boundary.

Proposition 2.6 (Density on the complete joint boundary)**.** *Points with $x, u \neq 0$ and exactly one vanishing F_i are dense in both $\partial\mathcal{D}_H$ and $\partial\mathcal{D}_L$. In particular regular joint edges are dense, including near all zero-transfer, zero-real-part, collinear and intersection strata.*

Proof. For the HT closure (2.9), $\sinh|x| \sinh|u| \leq 1$ by (2.5). For fixed real-part magnitudes a, b with product strictly below one, its angular fiber is

$$|y| \leq \arccos r_1(a, b), \qquad |v| \leq \arccos r_2(a, b).$$

At product one both intervals collapse to zero. At any boundary point at least one inequality is active. Approximate its magnitudes by $a_j, b_j > 0$ with $\sinh a_j \sinh b_j < 1$. If the first face is active, choose $|y_j| = \arccos r_1(a_j, b_j)$ with $y_j \to y$, and choose v_j strictly inside its second interval with $v_j \to v$. Preserve each nonzero real-part sign, and choose either sign at a zero real part. This gives regular points $F_1 = 0, F_2 > 0$ approaching the prescribed point. Exchange the faces when the second is active. The construction also covers strip endpoints and collapsed fibers.

A point of the LT boundary has $\sinh x \sinh u \leq 1$, since a larger product would make both inequalities strict. It has an active cosine inequality: $x = 0$ forces the second active and $u = 0$ the first. Choose $x_j, u_j > 0$ with product below one approaching the given point. For an active first face choose

$$|\cos y_j| = r_1(x_j, u_j), \quad y_j \to y, \qquad |\cos v_j| < r_2(x_j, u_j), \quad v_j \to v.$$

Continuity of the local cosine intervals permits these choices also when an endpoint tends to a cosine zero or to a multiple of π. They give $F_1 = 0, F_2 < 0$. Exchange the faces for the other case. □

Proposition 2.7 (Independent holomorphic edge coordinates)**.** *Where $S \neq 0$, put $C = A/S$ and $B = T/S$. Then*

$$\det \frac{\partial(C, B)}{\partial(z, w)} = \frac{p^2 - 1}{S^3}. \tag{2.16}$$

At a regular edge these are local holomorphic coordinates and $\Im B \neq 0$. At an edge with $b = \epsilon$, the coordinate $h = C - \epsilon B$ has boundary $\Im h = 0$, $\Re h \in (-1, 1)$; the physical component is the entire local complementary side. For $a = \epsilon$, exchange transfer axes. On every regular joint boundary patch, points whose free coefficient is $\cos(2\pi k/q)$, with q an odd prime and $1 \leq k \leq (q-1)/2$, are dense.

Proof. The derivatives

$$C_z = -\frac{\cosh w}{S^2}, \quad C_w = \frac{\cosh z\, T}{S}, \quad B_z = -\frac{T \cosh z}{S^2}, \quad B_w = \frac{\cosh w}{S}$$

give (2.16); also $D = |S|^2 \Im B$. In these coordinates the spectral condition is the parallelogram $C \in [-1, 1] + B[-1, 1]$. Near the interior of its $b = \epsilon$ edge,

$$b - \epsilon = \frac{\Im(C - \epsilon B)}{\Im B},$$

while the other coefficient remains strictly interior. The two local sides are therefore the spectral side and one connected complementary side. The latter meets the physical component and lies entirely in it. Holding B fixed makes h range over a real interval. Prime-grid angles have arbitrarily small mesh, so their cosines are dense there. □

On an HT F_1 face the endpoint is $b = \operatorname{sign} u$; on an LT F_1 face it is $b = \operatorname{sign}(\cos y)$. On F_2 faces exchange x, y with u, v. Both endpoint signs must consequently be retained in the analytic boundary theorem.

2.3 Every positive-ratio slice

For $\lambda > 0$ define

$$P_\lambda(a) = \cosh a \tanh(\lambda a), \quad Q_\lambda(a) = \cosh(\lambda a) \tanh a, \quad Y_\lambda = \frac{\pi}{2 \max(1, \lambda)}.$$

These functions are odd analytic bijections of $\mathbb{R}$ with strictly positive derivatives. Their restrictions to $[0, \infty)$ thus have increasing inverses. Write

$$H_\lambda(b) = \min\{P_\lambda^{-1}(\cos b), Q_\lambda^{-1}(\cos(\lambda b))\}, \quad |b| < Y_\lambda,$$

$$L_\lambda(b) = \max\{P_\lambda^{-1}(|\cos b|), Q_\lambda^{-1}(|\cos(\lambda b)|)\}.$$

Theorem 2.8 (Exact pullbacks)**.** *For $\iota_\lambda(\zeta) = (\zeta/2, \lambda\zeta/2)$,*

$$\iota_\lambda^{-1}(\mathcal{D}_H) = \Omega_{H,\lambda} = \{a + ib : |b| < Y_\lambda,\ |a| < H_\lambda(b)\}, \tag{2.17}$$

$$\iota_\lambda^{-1}(\mathcal{D}_L) = \Omega_{L,\lambda} = \{a + ib : a > L_\lambda(b)\}. \tag{2.18}$$

These domains are simply connected. If $z_c > 0$ is the unique solution of $\sinh z_c \sinh(\lambda z_c) = 1$, then $P_\lambda(z_c) = Q_\lambda(z_c) = 1$ and $0 \le L_\lambda \le z_c$. The HT profile is positive in its interval and tends to zero at both ends.

Proof. For example

$$P_\lambda'(a) = \sinh a \tanh(\lambda a) + \lambda \cosh a \operatorname{sech}^2(\lambda a) > 0;$$

the exchanged derivative has the same property. Substitute $(x, y, u, v) = (a, b, \lambda a, \lambda b)$ into the exact component inequalities and invert these two increasing functions. This gives both pullbacks. The identities (2.5) give the claims about z_c, L_λ. At an HT endpoint at least one cosine vanishes, proving the limit of H_λ. The HT domain is star shaped by dilation. The LT map $(a, b) \mapsto (a - L_\lambda(b), b)$ is a homeomorphism to $(0, \infty) \times \mathbb{R}$. □

Proposition 2.9 (Regular points on each fixed-ratio boundary)**.** *For every $\lambda > 0$, $\lambda \neq 1$, all but a locally finite subset of either physical slice boundary are regular joint edges, and their edge maps are nonramified on the slice. The selected prime cosines are dense on every remaining regular arc and hence on both entire slice boundaries.*

Proof. Use the signed real-analytic profiles

$$R(b) = P_\lambda^{-1}(\cos b), \qquad U(b) = Q_\lambda^{-1}(\cos(\lambda b)), \qquad E_\lambda(b) = R(b)^2 - U(b)^2.$$

Here the inverses are those of the odd bijections on all of $\mathbb{R}$. Thus E_λ is analytic on all of $\mathbb{R}$, including every simultaneous cosine zero. Near zero,

$$\begin{aligned} E_\lambda(b) &= z_c \left(\frac{\lambda^2}{Q'_\lambda(z_c)} - \frac{1}{P'_\lambda(z_c)} \right) b^2 + O(b^4) \\ &= z_c \{ \lambda \coth(\lambda z_c) - \coth z_c \} b^2 + O(b^4). \end{aligned}$$

For the second equality put $h = \sinh z_c$, so $\sinh(\lambda z_c) = h^{-1}$, $P'_\lambda(z_c) = h(1+\lambda h)/\cosh z_c$, and $Q'_\lambda(z_c) = (1+\lambda h)/(h \cosh z_c)$. The displayed coefficient is nonzero when $\lambda \neq 1$, because $t \coth t$ is strictly increasing on $t > 0$; its derivative has positive numerator $\sinh t \cosh t - t$. Hence the zeros of E_λ are locally finite.

Except at these profile contacts and the two HT tips, a boundary point has $a \neq 0$ and exactly one vanishing F_i. The strict-face identity and (2.15) give all joint regularity conditions. There is no separate rational-ratio exception: all such contacts are already zeros of the same analytic function.

The edge map on an F_1 face is

$$g_\epsilon(\zeta) = \frac{A - \epsilon T}{S} = \frac{r_\epsilon + r_\epsilon^{-1}}{2}, \qquad r_\epsilon = e^{\epsilon \lambda \zeta} \tanh(\zeta/2). \tag{2.19}$$

At that face,

$$\frac{r'_\epsilon}{r_\epsilon} = \epsilon\lambda + \frac{1}{S}, \qquad \Re \frac{r'_\epsilon}{r_\epsilon} = \epsilon \left(\lambda + \frac{\sinh a \sinh(\lambda a) \cosh(\lambda a)}{\cosh a} \right) \neq 0.$$

The expression in parentheses is positive also when $a < 0$. Strictness of the free coefficient implies $r_\epsilon \neq \pm 1$, so $g'_\epsilon \neq 0$. On an F_2 face use $r = e^{\epsilon\zeta} \tanh(\lambda\zeta/2)$; there

$$\Re \frac{r'}{r} = \epsilon \left(1 + \lambda \frac{\sinh a \sinh(\lambda a) \cosh a}{\cosh(\lambda a)} \right) \neq 0.$$

Consequently each regular boundary arc maps biholomorphically to an interval of free coefficients, in which the prime cosines are dense. The two boundaries are continuous graphs or a continuous lens boundary and have no isolated points; deleting their locally finite exceptional subsets leaves a dense set. □

At a regular selected slice point $\zeta_0 \neq 0$, the map $(\zeta, \lambda) \mapsto (z, w) = (\zeta, \lambda\zeta)$ has determinant ζ_0. Composing with the edge chart and using the just-proved nonzero edge derivative gives $B = B(t, \lambda)$ by the holomorphic implicit-function theorem on a product polydisk, with t the nonramified edge coordinate. Shrinking a slightly larger product polydisk preserves the sign of $\Im B$ and gives common Cauchy bounds. The local analytic boundary estimates thus apply to the variable-B slices and their complex parameter neighborhoods.

2.4 The local collar and propagation to exceptional points

Lemma 2.10 (Inward-map collar)**.** *Let D be $\mathcal{D}_H$, $\mathcal{D}_L$, or either of their positive-ratio pullbacks. If $P \in \partial D$ and U is an open neighborhood of P, there is a neighborhood $V \subset U$ of P such that $D \cap V$ lies in one connected component of $D \cap U$. Every component of $D \cap U$ accumulating at P contains $D \cap V$.*

Proof. For HT use $H_s(Q) = (1-s)Q$; for LT in (x, u, y, v) coordinates use $H_s(Q) = Q + s(1, 1, 0, 0)$. These maps preserve D for $s \geq 0$ sufficiently small and move every boundary point strictly inside for

$s > 0$, by the closure arguments above. On a slice use the same HT dilation and the LT translation $\zeta \mapsto \zeta + s$, corresponding to the real direction $(1, \lambda, 0, 0)$.

Choose a small fixed $s > 0$ so that $H_t(P) \in U$ for $0 \le t \le s$, and choose a ball around $H_s(P)$ contained in $D \cap U$. By continuity and compactness of $[0, s]$, after shrinking V, every path $H_t(Q)$, $Q \in D \cap V$, stays in $D \cap U$ and its endpoint belongs to that ball. The paths and straight segments inside the ball join any two such Q. An accumulating component meets $D \cap V$ and hence is this same component. □

Definition 2.11. For $f \in \mathcal{O}(D)$, a local holomorphic extension through $P \in \partial D$ is a function holomorphic on a neighborhood U of P that agrees with f on a connected component of $D \cap U$ accumulating at P.

Corollary 2.12 (Dense obstructions imply every componentwise obstruction)**.** *For any domain D in the preceding lemma, suppose $f \in \mathcal{O}(D)$ has no holomorphic extension agreeing on the full local domain side at a dense subset of ∂D. Then no neighborhood of any $P \in \partial D$ admits a holomorphic function agreeing with f on an accumulating connected component of its intersection with D.*

Proof. An extension agreeing on such a component agrees on $D \cap V$ by the collar lemma. Choose a dense obstructing point in V and restrict the extension to a neighborhood contained in V. □

2.5 The isotropic disk and exterior covering

Lemma 2.13 (Exact isotropic geometry)**.** *At $\lambda = 1$, with $s = \sinh\zeta$,*

$$\iota_1^{-1}(\Sigma) = \{|\sinh\zeta| = 1\}, \quad \Omega_{H,1} = \{|b| < \pi/2,\ |a| < \operatorname{arsinh}(\cos b)\}, \quad \Omega_{L,1} = \{a > \operatorname{arsinh}(|\cos b|)\}.$$

The map sinh *is a biholomorphism from $\Omega_{H,1}$ onto the unit disk and a universal covering from $\Omega_{L,1}$ onto the unit-circle exterior, with deck translations $\zeta \mapsto \zeta + 2\pi i k$.*

Proof. Here $S = T = s$ and $A = 1 + s^2$. For $s \neq 0$ the unsquared condition is $s + s^{-1} \in [-2, 2]$, equivalent to $|s| = 1$; $s = 0$ is not spectral. Since $P_1 = Q_1 = \sinh$, the exact profiles follow. The identity $|\sinh(a + ib)|^2 = \sinh^2 a + \sin^2 b$ maps the bounded HT lens into the disk and its boundary into the circle. The map is injective on $|\Im\zeta| < \pi/2$: equal hyperbolic sines require $\zeta_1 = \zeta_2 + 2\pi i k$ or $\zeta_1 = -\zeta_2 + (2k+1)\pi i$, and only the first with $k = 0$ is possible. Its derivative has no zero there. Its image is nonempty, open and relatively closed in the disk by the boundary identity and boundedness, hence is the whole disk.

For the exterior put $\omega = e^{\zeta}$. The equation $\omega - \omega^{-1} = 2s$ has exactly one root with $|\omega| > 1$ when $|s| > 1$: the root moduli have product one and a unit root would give $s \in i[-1, 1]$. The outside root is holomorphic, using $\sqrt{s^2 + 1} = s\sqrt{1 + s^{-2}}$ with its binomial branch. Its image is

$$\{re^{ib} : r > \exp(\operatorname{arsinh}|\cos b|)\},$$

by the same modulus identity. This root map is a biholomorphism. Its full exponential preimage is the simply connected LT epigraph, so exponentiation supplies the universal covering and the stated deck group. The reduced susceptibility requires its own termwise deck invariance to descend; inverse temperature remains on this cover. □

3 The physical observable and joint holomorphy

Throughout this paper $z = 2K_x$ and $w = 2K_y$ are independent complex coordinates. Write

$$S = \sinh z, \qquad T = \sinh w, \qquad A = \cosh z \cosh w, \qquad p = ST.$$

For real positive K_x, K_y, write η for the dimensionless uniform field. The zero-field responses are

$$\mathcal{X}_H(K_x, K_y) = \sum_{(m,l)\in\mathbb{Z}^2} \langle \sigma_{0,0}\sigma_{m,l}\rangle, \qquad 0 < p < 1, \tag{3.1}$$

$$\mathcal{X}_L(K_x, K_y) = \sum_{(m,l)\in\mathbb{Z}^2} \big(\langle \sigma_{0,0}\sigma_{m,l}\rangle_+ - M^2\big), \qquad p > 1, \tag{3.2}$$

$$M^2 = (1 - p^{-2})^{1/4}.$$

The high-temperature state is the unique Gibbs state. The subscript + means the infinite-volume plus state; the fourth root in the last line is positive. These definitions include the origin, both axes and all quadrants. The low-temperature onsite term is $1 - M^2$.

3.1 Real-temperature form factors and state identification

Only the real, noncritical spin form factors are imported from the exact solution. The relevant finite-cylinder formulas at arbitrary separation are [1, (35)–(39)]; their finite-lattice matrix-element factorization is proved in [2, Theorem 6.1], with the paramagnetic vacuum change in its Remark 6.2. In our convention put

$$B = T/S > 0, \quad C = A/S, \quad \cosh\gamma(q) = C - B\cos q, \quad \gamma(q) > 0.$$

Then the infinite-plane coefficient is

$$f_N(m,l) = \frac{B^{N^2/2}}{N!} \int_{\mathbb{T}^N} \prod_{i=1}^{N} \frac{e^{-|m|\gamma(q_i)+\mathrm{i}lq_i}}{\sinh\gamma(q_i)} \prod_{i<j} \left\{ \frac{\sin((q_i - q_j)/2)}{\sinh((\gamma(q_i) + \gamma(q_j))/2)} \right\}^2 \prod_{i=1}^{N} \frac{\mathrm{d}q_i}{2\pi}. \tag{3.3}$$

With positive real square roots the two correlations are

$$\langle \sigma_0\sigma_{m,l}\rangle = \xi_H \sum_{\substack{N\geq 1 \\ N \text{ odd}}} f_N(m,l), \qquad \xi_H = (1-p^2)^{1/4}p^{-1/2}, \tag{3.4}$$

$$\langle \sigma_0\sigma_{m,l}\rangle_+ = \xi_L \left(1 + \sum_{\substack{N\geq 2 \\ N \text{ even}}} f_N(m,l) \right), \qquad \xi_L = (1-p^{-2})^{1/4} = M^2. \tag{3.5}$$

This is the normalization of [3, (1.8)–(1.10)]. In particular the anisotropy exponent in (3.3) is $N^2/2$.

On a real noncritical compact set the dispersion has a uniform positive gap and is analytic in an angular strip. finite-circumference corrections $\eta, \Lambda^{-1}, \log\xi_T$ in [1] tend uniformly to zero, by the exponentially convergent trapezoidal rule applied to their sum-minus-integral expressions. If L is the circumference, the unnormalized Ramond and Neveu–Schwarz sums of a strip-analytic function differ by $O(Le^{-hL})$: their Fourier expansions select the multiples of L, with respective factors 1 and $(-1)^k$. Apply this to γ and to $\log(1 - e^{-\gamma(q)-\gamma(q')})$. The mixed double-grid difference of $\log\sinh((\gamma(q) + \gamma(q'))/2)$ in [1, (25)] gives

$$|\Lambda_L^{-1}| + \sup_q |\eta_L(q)| \leq CLe^{-hL}, \qquad |\log\xi_{T,L}| \leq CL^2e^{-2hL}.$$

Thus at each fixed separation the entire correction is bounded by $C_{m,l}C^N$, uniformly in L, N. Terms with $N > L$ vanish on the discrete grid. For the real roots $0 < \rho(q) = e^{-\gamma(q)} \le r < 1$,

$$\left| \frac{\rho(q) - \rho(q')}{1 - \rho(q)\rho(q')} \right| \le r < 1.$$

The matching argument in Lemma 4.1 is an algebraic identity on discrete angular grids as well as on continuous circles. It gives a common bound $\exp\{-cN^2 + O(N \log(N+1))\}$ at fixed separation, including the finite-size one-body corrections and factorial normalization. Consequently the particle sum passes through the cylinder limit; each fixed-index Riemann sum gives (3.3).

Every local cylinder limit is a translation-invariant and spin-flip invariant Gibbs state. The Gibbs-state theorem [5, Theorem 1] applies with the vertex-transitive group chosen to be translations: the two unequal positive nearest-neighbor couplings satisfy its nonnegativity, invariance, irreducibility and summability hypotheses. That state is the equal mixture of the plus and minus states. Its two-spin correlations equal those of the plus state. This proves the state identification in (3.5); subtracting $M^2 = \xi_L$ removes exactly its zero-particle term. No response of a zero-mean mixed state is substituted for (3.2).

3.2 Double contours and the full lattice

Let ζ, η now denote independent contour variables, and set

$$\mathcal{D}(\zeta, \eta; z, w) = A - \frac{S}{2}(\zeta + \zeta^{-1}) - \frac{T}{2}(\eta + \eta^{-1}), \qquad a(v, v') = \frac{v - v'}{1 - vv'}.$$

If $X = \prod_i \zeta_i$ and $Y = \prod_i \eta_i$, the same coefficient has the symmetric representation

$$f_N(m, l) = \frac{p^{N/2}}{N!} \oint \cdots \oint X^{|m|-1} Y^{|l|-1} \prod_{i<j} a(\zeta_i, \zeta_j) a(\eta_i, \eta_j) \prod_{i=1}^{N} \frac{\mathrm{d}\zeta_i \, \mathrm{d}\eta_i}{(2\pi \mathrm{i})^2 \mathcal{D}(\zeta_i, \eta_i)}. \tag{3.6}$$

The circles have radii strictly less than and sufficiently close to one. On the real physical region, at $\eta = e^{\mathrm{i}q}$ the inside root is $\zeta = \rho = e^{-\gamma(q)}$, and

$$\mathcal{D}_\zeta(\rho, \eta) = \frac{S(1 - \rho^2)}{2\rho^2}, \qquad \frac{\sqrt{p}}{\mathcal{D}_\zeta(\rho, \eta)} = \sqrt{B} \, \frac{\rho}{\sinh \gamma}.$$

Subtracting two dispersion equations gives the exact canceled pair

$$a(\rho_i, \rho_j) a(\eta_i, \eta_j) = -B \frac{(\eta_i - \eta_j)^2 \rho_i \rho_j}{\eta_i \eta_j (1 - \rho_i \rho_j)^2} = B \left\{ \frac{\sin((q_i - q_j)/2)}{\sinh((\gamma_i + \gamma_j)/2)} \right\}^2 =: P_{ij}. \tag{3.7}$$

The residues therefore give $B^{N/2}$, and the pairs give $B^{N(N-1)/2}$. This proves the full exponent and the normalization in (3.3). Each spatial coordinate can be reflected independently at fixed couplings.

The exact product of the two spatial geometric series is

$$\sum_{m,l \in \mathbb{Z}} X^{|m|-1} Y^{|l|-1} = \frac{(1 + X)(1 + Y)}{XY(1 - X)(1 - Y)} = \frac{1}{XY} + \frac{2(X^{-1} + Y^{-1})}{(1 - X)(1 - Y)}. \tag{3.8}$$

The first term is the onsite coefficient. Denoting by T_N the integral with the second numerator but without its factor two gives $C_N = f_N(0,0) + 2T_N$, where C_N sums all sites. The angular form of this complete coefficient is

$$C_N = \frac{1}{N!}\int_{\sum_i q_i = 0 \pmod{2\pi}} G_N(q)\frac{1+Z}{1-Z}\prod_{i=1}^{N-1}\frac{\mathrm{d}q_i}{2\pi}, \qquad G_N = B^{N/2}\prod_i \frac{1}{\sinh\gamma_i}\prod_{i<j} P_{ij}, \quad Z = \prod_i \rho_i. \tag{3.9}$$

The $N-1$ independent normalized measures correspond to $q_1, \ldots, q_{N-1}$, with $q_N = -\sum_{i<N} q_i$ modulo 2π.

Lemma 3.1 (Scalar matching identity)**.** *For even N,*

$$\operatorname{Pf}\left[\frac{v_i - v_j}{1 - v_i v_j}\right]_{i,j=1}^{N} = \prod_{i<j}\frac{v_i - v_j}{1 - v_i v_j}.$$

For odd N, adjoin $v_{N+1} = -1$; its upper-column entries are the removable constant one and the product on the right is the product over the original indices.

Proof. Set $a_i = (1+v_i)/(1-v_i)$, so the entries become $(a_i - a_j)/(a_i + a_j)$. As rational functions of a_N, the Pfaffian expansion and the product have the same residues at every $a_N = -a_i$, by induction on N. Both are bounded at infinity. Their difference is therefore constant in a_N. At $a_N = a_{N-1}$ both vanish, proving the induction; $N = 2$ starts it. The resulting rational identity includes all removable coincidences. Substitution of the ghost variable proves the odd formula. □

Proposition 3.2 (Joint normal convergence)**.** *On each of the two domains $\mathcal{D}_H, \mathcal{D}_L$ specified by the independent spectral geometry, the physical germ has a single-valued jointly holomorphic continuation. It is given by*

$$\mathcal{X}_H = (1-p^2)^{1/4}\sum_{\substack{N\geq 1\\ N \text{ odd}}} p^{(N-1)/2} F_N, \tag{3.10}$$

$$\mathcal{X}_L = (1-p^{-2})^{1/4}\sum_{\substack{N\geq 2\\ N \text{ even}}} p^{N/2} F_N, \tag{3.11}$$

where F_N is the integral (3.6) *with its $p^{N/2}$ factor removed and with the multiplier* (3.8)*. The sums converge normally, with every fixed mixed complex derivative, on compact subsets. The fourth roots are the binomial branches supplied by $|p| < 1$ on $\mathcal{D}_H$ and $|p| > 1$ on $\mathcal{D}_L$.*

Proof. On a compact parameter neighborhood in $\mathbb{C}^2 \setminus \Sigma$, the polynomial $\mathcal{D}$ has a nonzero minimum modulus on the product of unit circles. By uniform continuity that gap persists on a common thin product annulus and a slightly larger parameter polydisk. Choose fixed radii $r_x, r_y < 1$ in the annulus. All Schur entries on these circles are bounded by a constant independent of N. For even N, the Schur Pfaffian identity and Hadamard's determinant inequality give

$$\left|\prod_{i<j} a(\zeta_i, \zeta_j)\right| = |\operatorname{Pf}(a(\zeta_i, \zeta_j))| \leq C^{N/2} N^{N/4}.$$

For odd N, adjoin a ghost variable -1; its entries are the removable constant one and the dimension becomes $N+1$. Apply the estimate separately to the two contour-variable families. The one-body

factors and contour lengths cost C^N. Also $1-X$ and $1-Y$ are bounded away from zero uniformly in N, whereas X^{-1}, Y^{-1} cost only C^N. Thus, after increasing constants,

$$|F_N| \le C_0 C_1^N \frac{N^{(N+1)/2}}{N!}. \tag{3.12}$$

The right side is summable after multiplication by any fixed exponential in N. Multivariate Cauchy estimates on the larger polydisk give the same assertion for every fixed mixed derivative. On overlaps, radii sufficiently close to one can be deformed into one another inside the common gap annulus; all pair and product denominators stay nonzero. The local integrals therefore glue. All powers of p within (3.10)–(3.11) are integers. The binomial fourth roots introduce no monodromy, and the series prove joint holomorphy, including the HT axes $S = 0$ or $T = 0$.

On a real physical compact set the same contours in (3.6) additionally give $r_x^{N|m|} r_y^{N|l|}$. Their double geometric sum is bounded uniformly in $N \ge 1$. Thus the lattice and particle sums are absolutely interchangeable, and (3.8) gives exactly (3.1)–(3.2), with the stated zero-particle subtraction. Equality on a real open physical set determines the joint germ: iterating the one-variable identity theorem over a real rectangle proves the required uniqueness. □

The canonical root is that of Lemma 2.3. On a patch with $\rho \neq 0$, use the local logarithm $\gamma = -\log \rho$.

The angular representation used below is identified with this global continuation coefficient by coefficient. Indeed $1/\sinh\gamma = 2\rho/(1-\rho^2)$ and the rational expression for P_{ij} in (3.7) contain no logarithm. After including the physical normalization, their one-body prefactors are

$$(1-p^2)^{1/4} \frac{p^{(N-1)/2}}{S^N} \quad (N \text{ odd}), \qquad (1-p^{-2})^{1/4} \frac{p^{N/2}}{S^N} \quad (N \text{ even}).$$

All powers here are integers. Hence the normalized angular coefficient is single valued and holomorphic wherever $S \neq 0$ in the corresponding physical domain; the real angular torus has uniform root and denominator gaps on compact subsets. The set $\mathcal{D}_H \setminus \{S = 0\}$ is connected: radial contraction enters a small polydisk without crossing $z = 0$, and that polydisk minus $z = 0$ is connected. The domain $\mathcal{D}_L$ already has $S \neq 0$. The identity theorem from the real physical rectangle therefore gives equality with the double-contour coefficients throughout these sets. Near a regular edge $S, T \neq 0$, this also fixes the compatible local odd square root $\sqrt{B} = \sqrt{p}/S$. Normal convergence of the identified series follows from Proposition 3.2.

Remark 3.3 (The decoupled-axis restriction)**.** On $K_y = 0$, the lattice splits into independent horizontal chains. For $t = \tanh K_x$ near zero its response is

$$1 + 2\sum_{m \ge 1} t^m = \frac{1+t}{1-t} = e^{2K_x}.$$

This agrees with (3.10): $p = 0$, only $N = 1$ survives, and $F_1 = (A-S)^{-1} = e^{2K_x}$. The restriction is entire, including at boundary points of the joint domain on this line. The slice theorem uses the obstructions along the positive-ratio temperature lines themselves.

3.3 Energy-field normalization

For fixed real $J_x, J_y > 0$ and physical temperature $(K_x, K_y) = (\beta J_x, \beta J_y)$, a field of energy units enters the Hamiltonian as $-h\sum_v \sigma_v$, and $\eta = \beta h$. Finite-volume differentiation gives $\chi = \beta \mathcal{X}$.

The same statement holds after taking the prescribed infinite-volume phase limit. To justify the latter without exchanging an uncontrolled field limit and infinite sum, use the GHS inequality for arbitrary nonnegative couplings and fields [6, Theorem 1.5]. Pinning removed spins by fields tending to $+\infty$ shows that plus-box covariances increase with the box. Their nonnegative sums increase to the finite sum proved above. GHS concavity gives, for the plus-box magnetization,

$$m_V(h) - m_V(0) \le h\, m_V'(0) \le \beta h\, \mathcal{X}, \qquad h \ge 0.$$

The plus-state limit yields $m'(0+) \le \beta\mathcal{X}$. Conversely turn on h only on a finite set E. The resulting bounded local change of measure has derivative $\beta \sum_{v\in E} \mathrm{Cov}_{+,0}(\sigma_0, \sigma_v)$. Ferromagnetic monotonicity bounds its magnetization above by the uniform-field magnetization, with equality at $h = 0$. Increasing E proves the reverse inequality. Thus $m'(0+) = \beta\mathcal{X}$. Spin-flip symmetry gives the two-sided HT derivative; in LT the derivative is the plus-phase right derivative (or its minus-phase counterpart). This statement is made on temperature lines only.

4 Regular spectral edges and transverse curves

We use the two complex dispersion parameters on a common local polydisk. On a regular edge, exchange the two lattice axes if necessary. In the local coordinates $C = A/S,\ B = T/S$ of the geometry section, one edge is $C - B = b$, $b = \cos\phi \in (-1, 1)$. Choose the side for which $\Im B_0 > 0$; conjugation handles the other side. Let $B(u)$ be any holomorphic function near zero with $B(0) = B_0$, and set

$$W(q; u) = b + u + B(u)(1 - \cos q). \tag{4.1}$$

Both a constant-B complex transversal in $\mathbb{C}^2$ and each nonramified physical temperature-line transversal have this form.

At $u = 0$, the inside root lies in the closed lower half disk, with $\rho(0) = e^{-\mathrm{i}\phi}$. Compactness gives

$$\Im\rho(q) \le -\delta < 0, \qquad |\rho(q)| \le 1.$$

There is no $W = \pm 1$ on the real angular circle. The roots, their chosen logarithms, the one-body factors, and the canceled pairs (3.7) are analytic on one fixed complex neighborhood in (u, B, q). On $u = \mathrm{i}\varepsilon$, $\varepsilon > 0$,

$$\Re\gamma(q; \mathrm{i}\varepsilon) \ge c\{\varepsilon + \mathrm{dist}(q, 0)^2\}. \tag{4.2}$$

Indeed $\Im W = \varepsilon + \Im B(\mathrm{i}\varepsilon)(1 - \cos q)$ has this lower bound; the identity $\Im W = \sinh(\Re\gamma)\sin(\Im\gamma)$ on the compact branch then gives (4.2). This is the complement side of the parallelogram edge and hence the side of the bordering physical component.

Lemma 4.1 (Strict contraction on a complex polydisk)**.** *There are $\kappa > 0$, $C < \infty$, and a fixed complex parameter polydisk about this edge such that, for every fixed mixed derivative ∂^ν in the parameters,*

$$|\partial^\nu G_N(q)| \le C_\nu \exp\{-\kappa N^2 + CN\log(N + 1)\} \tag{4.3}$$

for all real angles and every $N \ge 1$.

Proof. The identity

$$|1 - vv'|^2 - |v - v'|^2 = (1 - |v|^2)(1 - |v'|^2) + 4\Im v\, \Im v'$$

gives a strict bound $|a(\rho_i, \rho_j)| \le q_0 < 1$. The denominator is separated from zero. By continuity a slightly larger constant below one still works on a common *complex* parameter polydisk, even at

parameters where individual root moduli exceed one. Every canceled pair P_{ij} remains bounded by C.

For even N, expand the Schur Pfaffian in the $\eta_i = e^{\mathrm{i}q_i}$ variables only. For each matching, combine its $N/2$ edges with the corresponding ρ factors using (3.7). The other $N(N-2)/2$ root factors retain their strict contraction. Thus

$$\left|\prod_{i<j} P_{ij}\right| \le (N-1)!!\, C^{N/2} q_0^{N(N-2)/2}.$$

For odd N, adjoining the constant ghost entry gives

$$\left|\prod_{i<j} P_{ij}\right| \le N!!\, C^{(N-1)/2} q_0^{(N-1)^2/2}.$$

These identities first hold away from reciprocal η pairs and extend by the analytic canceled formula. There is no division by a root-Schur zero. One-body factors cost C^N. Cauchy's inequalities on the larger polydisk prove all the derivative bounds. They are applied to G_N, never across a pole of $(1+Z)/(1-Z)$. □

Lemma 4.2 (One fixed collision block for the entire mixed-derivative tail)**.** *Fix $\phi = 2\pi a/\ell \in (0,\pi)$, where ℓ is an odd prime. Let $\mathcal{P} \Subset \mathcal{P}^+$ be concentric complex parameter polydisks. Suppose $B(u,\eta)$ is holomorphic on $\{|u| < R\} \times \mathcal{P}^+$, with*

$$|B(u,\eta)| \le M, \qquad \Im B(0,\eta) \ge \beta > 0 \quad (\eta \in \mathcal{P}^+).$$

Use $W(q;u,\eta) = \cos\phi + u + B(u,\eta)(1-\cos q)$ and the common branch $\gamma(0,0,\eta) = \mathrm{i}\phi$ in the complete coefficient $C_N(u,\eta)$ of (3.9). For odd parity put $n = \ell$, $k = (n^2-1)/2$; for even parity put $n = 2\ell$, $k = n^2/2 - 1$. There are $\epsilon_0, \kappa > 0$ such that, for every multi-index α and every $0 \le j \le k$,

$$\begin{gathered}\sup_{\eta\in\mathcal{P}} |\partial_u^j \partial_\eta^\alpha C_N(\mathrm{i}\epsilon,\eta)| \le \frac{C_{n,j,\alpha} N^j}{N!} \exp\{-\kappa N^2 + C_n N \log(N+1)\}, \\ 0 < \epsilon < \epsilon_0, \quad N > n, \quad N \equiv n \pmod 2.\end{gathered} \tag{4.4}$$

In particular the entire higher-index tail and all these mixed derivatives are bounded and have locally uniform boundary limits as $\epsilon \downarrow 0$. Every lower index of the chosen parity is jointly holomorphic on a neighborhood of $\{0\} \times \overline{\mathcal{P}}$.

The constants can be chosen uniformly for all B with the displayed common domain and bounds. They may depend on the fixed phase, R, M, β, the two parameter polydisks, and the indicated derivative orders. Taking $B(u,\eta) = B_0 + \eta$ includes derivatives in the independent dispersion variables (u, B).

Proof. Cauchy's inequality in u, followed by a uniform shrinking of its disk, gives $\Im B(u,\eta) \ge \beta/2$ for $|u| < r$ and $\eta \in \mathcal{P}^+$. At $u = 0$, the corresponding compact set of roots is contained in

$$\{\rho : |\rho| \le 1,\ \Im\rho \le -\delta\}$$

for some $\delta > 0$. Indeed B ranges in a bounded closed subset of the upper half-plane, q ranges over a compact circle, and $W = \pm 1$ never occurs. The chosen lower-half-plane root, its logarithm, and all one-body and canceled-pair factors consequently extend to a common complex u disk, uniformly

in η. On a slightly smaller such disk the strict root-Schur estimate in Lemma 4.1 holds with one constant $\theta < 1$, even where individual root moduli exceed one.

Fix an integer $m \geq 1$ and define the parameter-independent factor

$$V_m(q) = \prod_{1 \leq a < b \leq m} \sin^2((q_a - q_b)/2).$$

For $N \geq m$, factor $G_N = V_m A_{N,m}$ algebraically. Namely, replace each pair internal to the first m labels by $B/\sinh^2((\gamma_a + \gamma_b)/2)$, leaving the other pairs and the one-body factors as they stand. This defines $A_{N,m}$ holomorphically also at the zeros of V_m. The internal factors are bounded by a constant depending on m; the $m(N-m)$ cross pairs cost at most $C^{m(N-m)}$. Apply the matched-Pfaffian estimate only to the remaining $N-m$ labels, using its even or ghost-augmented odd version as appropriate. The empty product is one. It follows that

$$|A_{N,m}| \leq C_m \exp\{-\kappa(N-m)^2 + C(N-m)\log(N-m+1) + C_m N\}.$$

Cauchy's inequality on the common larger u disk therefore gives, for each fixed $r \geq 0$,

$$|\partial_u^r G_N(q; u, \eta)| \leq C_{m,r} V_m(q) \exp\{-\kappa N^2 + C_m N \log(N+1)\}, \tag{4.5}$$

uniformly for real angles and $\eta \in \mathcal{P}^+$. The product kernel is estimated on the inward approach below.

On $u = \mathrm{i}\epsilon$ let

$$Q_0(q) = \sum_{a=1}^{N} \operatorname{dist}(q_a, 2\pi\mathbb{Z})^2.$$

The common positive imaginary margin gives $\Re\gamma(q; \mathrm{i}\epsilon, \eta) \geq c(\epsilon + \operatorname{dist}(q, 2\pi\mathbb{Z})^2)$. Hence, with $Z = \exp(-\sum_a \gamma_a)$,

$$|1 - Z| \geq 1 - |Z| \geq c \min\{1, N\epsilon + Q_0\}.$$

All fixed u derivatives of each γ_a are uniformly bounded. The chain rule for the exponential and the rational kernel thus yields

$$\left|\partial_u^r \frac{1+Z}{1-Z}\right| \leq C_j N^j (1 + Q_0^{-j-1}), \qquad 0 \leq r \leq j, \tag{4.6}$$

uniformly in $\eta \in \mathcal{P}^+$ and $0 < \epsilon < \epsilon_0$. For example, derivatives of $\coth(\Gamma/2)$ of order s are bounded by $C_s(1 + (\Re\Gamma)^{-s-1})$ on $\Re\Gamma > 0$; each derivative of $\Gamma = \sum_a \gamma_a$ costs at most $C_j N$.

Now take $m = n + 1$. Every larger same-parity index satisfies $N \geq m + 1$, so $q_1, \ldots, q_m$ are independent variables of the constrained torus integral. Choose these representatives in $[-\pi, \pi]$ and set $R_m^2 = \sum_{a \leq m} q_a^2$. Then $Q_0 \geq R_m^2$ and $V_m \leq C_m R_m^{m(m-1)}$. The fixed-dimensional integral

$$\int_{[-\pi,\pi]^m} V_m(q)(1 + R_m^{-2j-2}) \prod_{a=1}^{m} \frac{\mathrm{d}q_a}{2\pi}$$

is finite: near zero its radial power is $R_m^{m^2-2j-3}\,\mathrm{d}R_m$, and $m^2 > 2j + 2$ for every $j \leq k$ in either parity. Every unused normalized angular measure has mass one. Applying the product rule to (3.9) and using (4.5)–(4.6) proves (4.4) with $\alpha = 0$, uniformly on the larger parameter polydisk.

For fixed $\epsilon > 0$, C_N and its u derivatives are holomorphic in η throughout $\mathcal{P}^+$: the same positive imaginary margin keeps the whole real torus on the physical side for every complex η there. Apply the multivariate Cauchy formula on parameter circles of a fixed radius between $\mathcal{P}$ and $\mathcal{P}^+$. It adds

only $\alpha!\prod_a d_a^{-\alpha_a}$ to the preceding bound and proves (4.4). Thus each fixed tangential derivative retains the collision powers on the smaller polydisk.

The majorant is summable in N. For each fixed higher index its integrand and all required derivatives have a pointwise boundary limit away from the all-zero collision, and the displayed fixed-dimensional majorant applies uniformly. Dominated convergence, first for each index and then for their summable series, proves the locally uniform boundary-limit assertion; the same parameter Cauchy formula gives every mixed limit. For a lower index, $|Z| = 1$ at $u = 0$ is possible only when every angle is zero, where $Z = e^{-\mathrm{i}N\phi} \neq 1$. Compactness of that fixed torus and the closed parameter polydisk gives a common denominator gap, which persists on a neighborhood of $u = 0$. Thus every lower coefficient is jointly holomorphic there. □

Proposition 4.3 (Nonzero full-sum edge obstruction on every holomorphic transversal)**.** *Under* (4.1) *and the rational-phase hypotheses of Lemma 4.2, the complete sum of the selected parity has no holomorphic extension through $u = 0$. The assertion is locally uniform in independent complex B parameters. It remains valid after multiplication by a holomorphic nonvanishing prefactor and after a nonramified change of transversal coordinate. It holds for both endpoint signs in the even case.*

Proof. Put $\alpha = (n^2 - 1)/2$ and

$$P_n = \prod_{r=1}^{n-1} \frac{r!}{2^r}.$$

The Gaussian Vandermonde period on the *independent-coordinate* measure is

$$\int_{q_n = -\sum_{i<n} q_i} e^{-Q} \Delta(q)^2 \, \mathrm{d}q_1 \cdots \mathrm{d}q_{n-1} = \frac{n!\, \pi^{(n-1)/2} P_n}{\sqrt{n}} =: \mathcal{G}_n. \tag{4.7}$$

To derive it, expand the ordinary n-variable Vandermonde determinant in monic Hermite polynomials. Their Gaussian norms are $\sqrt{\pi}\, r!/2^r$, so that integral is $n!\pi^{n/2}P_n$. Setting $x_i = q_i + m$, $\sum q_i = 0$, has Jacobian n, and the m integral is $\sqrt{\pi/n}$. Division proves (4.7).

On the unique collision chart,

$$\gamma(q; u) = \mathrm{i}\phi - \frac{\mathrm{i}}{\sin\phi}\left(u + \frac{B_0 q^2}{2}\right) + O(u^2 + uq^2 + q^4).$$

The leading singular integral for C_n is consequently

$$\frac{2\mathrm{i}^{1-n^2} B_0^{n^2/2}}{n!(2\pi)^{n-1} 2^{n(n-1)} \sin^{n^2-1}\phi} \int \frac{\Delta(q)^2}{nu + (B_0/2)Q} \, \mathrm{d}q_1 \cdots \mathrm{d}q_{n-1}. \tag{4.8}$$

For odd n the local square root of B is the one used in the physical angular coefficient. Its sign can change the displayed amplitude's sign but cannot make it zero. For even n the power in the original integrand is an integer; the square root below is fixed by the Gaussian period.

After k derivatives, the model integral is evaluated from

$$\int \frac{\Delta(q)^2}{(a + bQ)^{k+1}} \, \mathrm{d}q = \mathcal{G}_n \, b^{-\alpha} \frac{\Gamma(k + 1 - \alpha)}{\Gamma(k + 1)} a^{\alpha - k - 1},$$

initially in a common right half-plane and then by analytic continuation to $a = nu$, $b = B_0/2$. More explicitly, rotate both coefficients by $-\mathrm{i}$: $a' = n\varepsilon > 0$ and $b' = -\mathrm{i}B_0/2$ has positive real part.

The absolutely convergent Laplace–Gaussian calculation for a', b' fixes the continuation and its square root using $\Re b' > 0$. For odd n, $k = \alpha$ is a multiple of four; for even n, $k = \alpha - \frac{1}{2}$ is odd. Substitution gives

$$\lim_{\varepsilon \downarrow 0} u\, \partial_u^k C_n(u) = \frac{2^{1-k}\sqrt{B_0}\, P_n n^{k-3/2}}{\pi^{(n-1)/2} \sin^{2k} \phi} \neq 0, \qquad u = \mathrm{i}\varepsilon, \quad n \text{ odd}, \tag{4.9}$$

$$\lim_{\varepsilon \downarrow 0} u^{1/2} \partial_u^k C_n(u) = -\mathrm{i}\, \frac{2^{1-\alpha}\sqrt{B_0}\, P_n n^{k-1}}{\pi^{(n-2)/2} \sin^{n^2-1} \phi} \neq 0, \qquad u = \mathrm{i}\varepsilon, \quad n \text{ even}. \tag{4.10}$$

In the second line $u^{1/2} = e^{\mathrm{i}\pi/4}\sqrt{\varepsilon}$.

To compare with the exact fixed-n coefficient, write its integrand on the collision chart as $\Delta^2 f(q,u)/H(q,u)$. It is even under simultaneous $q \mapsto -q$, and, with $c = -\mathrm{i}/\sin\phi$,

$$H = c\{nu + (B_0/2)Q\} + O((|u| + Q)^2), \quad H_u = cn + O(|u| + Q), \quad f = f_0 + O(|u| + Q).$$

Both the exact and model denominators have modulus at least $c_1(\varepsilon + Q)$ on $u = \mathrm{i}\varepsilon$. The top derivative has leading term $(-1)^k k! f H_u^k / H^{k+1}$; all other terms have denominator power at most k. Subtracting the constant quadratic model therefore gives the pointwise bound

$$|\partial_u^k(\text{exact integrand} - \text{model integrand})| \le C\Delta^2(\varepsilon + Q)^{-k}.$$

Its radial integral is $O(\log(1/\varepsilon))$ for odd n, where $k = \alpha$, and $O(1)$ for even n, where $k = \alpha - 1/2$. Extending the differentiated model from the chart to the whole tangent plane costs $O(1)$, since its radial tails are R^{-3} and R^{-2}, respectively. Outside the chart the exact denominator has a gap. The variation of $B(u)$ contributes $O(uQ)$ to the denominator remainder, so these estimates give both limits for the exact coefficient.

Lemma 4.2 bounds all higher derivatives in the entire tail, and lower indices are analytic. Lower derivatives of the even first term are bounded; the next lower odd derivative may be logarithmic, but all are $o(u^{-1})$. Thus the nonzero top amplitudes cannot be canceled in the full sum or by product-rule terms from an analytic nonvanishing physical prefactor. The chain rule proves the statement for every nonramified curve parameter.

At the negative endpoint replace q_i by $\pi + q_i'$. For even N the total momentum constraint is unchanged and $B^{N^2/2} = (-B)^{N^2/2}$. Apply the preceding proof with $-B$, conjugating if necessary to have positive imaginary part. Exchanging transfer axes handles the other two edges. For odd N this shift instead makes the total reduced angle an odd multiple of π; that distinct case is addressed by the full boundary-jet theorem in the next section. □

Uniformity of the first amplitude in holomorphic families. Under the common-family hypotheses of Lemma 4.2, all branch neighborhoods, scalar derivative bounds, numerator factorizations, and quadratic-denominator bounds in the fixed-n calculation can be chosen uniformly on $\mathcal{P}^+$. The variation $[B(u,\eta) - B(0,\eta)]Q = O(|u|Q)$ enters the denominator remainder. The comparison errors are therefore uniformly $O(1 + |\log\epsilon|)$ in the odd case and $O(1)$ in the even case. The nonzero limits (4.9)–(4.10) hold locally uniformly in η, with B_0 replaced by $B(0,\eta)$ and its compatible common square root. Multivariate Cauchy on $\mathcal{P}^+$ then gives these limits after every fixed η derivative. A derivative of the leading coefficient can vanish, while the undifferentiated coefficient is nonzero for each member of the family. The same argument applies to the lower derivatives and to an analytic nonvanishing physical prefactor. Thus the full-sum obstruction and its mixed estimates hold uniformly on each compact admissible holomorphic family.

At a negative endpoint of the even sum, the change $q_a = \pi + q'_a$ preserves the momentum constraint and replaces B by $-B$; the equality $B^{N^2/2} = (-B)^{N^2/2}$ holds for even N. The preceding family argument therefore applies after this change and, when necessary, complex conjugation of all data. The conjugated family is understood holomorphically as $\overline{B(\bar{u}, \bar{\eta})}$ on the reflected polydisk.

5 The full odd sum at a regular negative spectral edge

Along a transverse analytic curve we use the holomorphic anisotropy with its exact particle-dependent power in the full constrained angular integral and sum over all odd orders.

5.1 Statement, branch, and the physical side

Fix

$$\phi = \frac{2\pi r}{p} \in (0, \pi), \qquad (r, p) = 1, \qquad p \geq 3 \text{ odd}, \qquad b = \cos\phi, \qquad s_\phi = \sin\phi > 0. \tag{5.1}$$

Let B be holomorphic on a disk about zero and suppose that

$$\Im B(0) < 0. \tag{5.2}$$

Shrink this disk so that $B \neq 0$ and $\Im B < 0$ throughout it, and fix one holomorphic logarithm of B. For $q = \pi + \delta$, define

$$W(t, \pi + \delta) = b + B(t)\big(t - 1 + \cos\delta\big), \qquad g(t, \delta) = \operatorname{arcosh} W(t, \pi + \delta), \qquad g(0, 0) = \mathrm{i}\phi. \tag{5.3}$$

At $t = 0$, W is in the open upper half-plane except at $\delta = 0 \bmod 2\pi$, where it equals $b \in (-1, 1)$. Consequently the branch in (5.3) extends to a fixed complex neighborhood of all real angles and a fixed parameter disk. It is even and 2π-periodic in δ. Put $\gamma(t, q) = g(t, q - \pi)$ and $\rho(t, q) = e^{-\gamma(t,q)}$. At $t = 0$, $|\rho| \leq 1$, and the compact set of these roots lies strictly in the lower half-plane:

$$\Im\rho(0, q) \leq -\delta_0 < 0. \tag{5.4}$$

The logarithm used for γ is fixed by $0 < \Im\gamma(0, q) < \pi$. In particular all the one-body denominators and the canceled pair denominators below are nonzero on a fixed complex neighborhood.

For odd $n \geq 1$, let

$$\mathcal{T}_n = \{(q_1, \dots, q_n) \in (\mathbb{R}/2\pi\mathbb{Z})^n : q_1 + \cdots + q_n = 0 \bmod 2\pi\}, \qquad q_n = -\sum_{i<n} q_i \bmod 2\pi.$$

The measure on this torus is $\mathrm{d}q_1 \cdots \mathrm{d}q_{n-1}$. Include the normalization in the numerator:

$$H_n(t, q) = \frac{B(t)^{n^2/2}}{n!(2\pi)^{n-1}} \prod_{i=1}^{n} \frac{1}{\sinh\gamma(t, q_i)} \prod_{i<k} \left[\frac{\sin((q_i - q_k)/2)}{\sinh((\gamma(t, q_i) + \gamma(t, q_k))/2)}\right]^2, \tag{5.5}$$

$$\Gamma_n(t, q) = \sum_{i=1}^{n} \gamma(t, q_i), \tag{5.6}$$

$$C_n(t) = \int_{\mathcal{T}_n} H_n(t, q) \coth\left(\frac{\Gamma_n(t, q)}{2}\right) \mathrm{d}q. \tag{5.7}$$

The power of B uses the chosen common logarithm. Changing its square-root sign changes every odd coefficient by the same sign.

The relevant open side of the boundary is

$$\mathcal{U}^- = \{\, |t| < r_0 :\ \Im(B(t)t) > 0 \,\}. \tag{5.8}$$

Here

$$\Im W(t, \pi + \delta) = \Im(B(t)t) - \Im B(t)(1 - \cos\delta) > 0$$

for every real angle, so the chosen roots satisfy $|\rho| < 1$. This is the local physical side when the regular negative endpoint has the lower-half-plane root $\rho(0, \pi) = e^{-\mathrm{i}\phi}$. It contains $t = -\epsilon$ for every sufficiently small $\epsilon > 0$. When these coordinates are obtained from $W(q) = C - B\cos q$, they are exactly

$$t = \frac{C + B - b}{B}.$$

The curve hypothesis is that this t is nonramified, with $B(t)$ the anisotropy along the curve. Complex conjugation gives the opposite orientation.

Theorem 5.1 (Universal analytic-curve obstruction at a negative edge)**.** *Under* (5.1)–(5.2)*, the series*

$$F(t) = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} C_n(t)$$

converges normally on $\mathcal{U}^-$. Every inward derivative has a finite limit

$$a_j = \lim_{\epsilon \downarrow 0} \frac{F^{(j)}(-\epsilon)}{j!}, \qquad j = 0, 1, 2, \ldots. \tag{5.9}$$

Nevertheless its formal boundary Taylor series has radius zero. More precisely, as n tends to infinity through odd multiples of p, there are integers j_n such that

$$j_n \sim n^2 \log n, \tag{5.10}$$

$$|a_{j_n}|^{1/j_n} \sim \frac{4 j_n}{e\pi^2 \log j_n}. \tag{5.11}$$

It follows that F has no holomorphic extension through $t = 0$ from $\mathcal{U}^-$. If P is holomorphic near zero and $P(0) \neq 0$, the same nonextension conclusion holds for PF.

The hypotheses and all estimates are locally uniform for complex holomorphic families $B(t, \eta)$: it suffices to work on a common parameter disk and a relatively compact parameter polydisk on which $\Im B(0, \eta) \leq -\beta < 0$ and B has a common holomorphic bound. The constants can depend on that disk, that bound, β, and the fixed rational phase. The selected integers may depend on the family parameter, $j_n = j_n(\eta)$.

We identify a single particle index that dominates the others in modulus at selected Taylor orders, allowing arbitrary phases for the other indices.

5.2 A uniform Gaussian numerator bound and all boundary jets

Write $\mathcal{A}(v, w) = (v - w)/(1 - vw)$ and $y_i = e^{\mathrm{i}q_i}$. Subtraction of two dispersion equations gives the canceled identity

$$\mathcal{A}(\rho_i, \rho_k)\mathcal{A}(y_i, y_k) = B(t) \left[\frac{\sin((q_i - q_k)/2)}{\sinh((\gamma_i + \gamma_k)/2)} \right]^2. \tag{5.12}$$

This is an identity of holomorphic canceled functions, including reciprocal y-pairs. The elementary identity

$$|1-vw|^2-|v-w|^2=(1-|v|^2)(1-|w|^2)+4\Im v\Im w \tag{5.13}$$

and (5.4) imply $|\mathcal{A}(\rho_i,\rho_k)|\le\theta<1$ at $t=0$. The same strict bound, with a slightly larger $\theta<1$, holds on a fixed complex parameter disk for all real angles. Every complete canceled pair in (5.12) is bounded there.

The Schur identity here is $\mathrm{Pf}(\mathcal{A}(y_i,y_k))_{i,k=1}^N=\prod_{i<k}\mathcal{A}(y_i,y_k)$ for even N. It follows by induction in N: as rational functions of y_N, the two sides have the same simple residues at $y_N=1/y_k$. In checking the residues, the other paired factors cancel by $\mathcal{A}(a,b)\mathcal{A}(a,1/b)=1$, and the remaining product is the induction hypothesis. Their difference is constant because both are bounded at infinity, and that constant is zero at $y_N=y_{N-1}$. The identity then extends from generic tuples by rational continuation.

Apply this identity only to the y-variables, adjoining the ghost variable -1 because n is odd. The entry $\mathcal{A}(y_i,-1)$ is the removable constant one. Each of the $n!!$ matchings has $(n-1)/2$ real edges. Combine each matched y-factor with the corresponding ρ-factor by (5.12). The remaining $(n-1)^2/2$ ρ-factors are at most θ in modulus. Thus, with the one-body factors and normalization included,

$$\sup_{|t|<r_0,\ q\in\mathcal{T}_n}|H_n(t,q)|\le E_n,\qquad E_n=\exp\{-\kappa n^2+Cn\log(n+1)+C\} \tag{5.14}$$

for some $\kappa>0$. Initially the Pfaffian calculation can be made off its apparent reciprocal-pair divisors; the canceled identity extends the estimate to those divisors. There is no division by a possibly zero ρ-factor.

Choose shortest lifts $q_i=\pi+\delta_i \bmod 2\pi$, with $\delta_i\in[-\pi,\pi]$, and put

$$S(q)=\sum_{i=1}^n\delta_i^2.$$

The odd momentum constraint says $\sum_i\delta_i=w\pi$, where w is odd. Therefore

$$S(q)\ge\frac{\pi^2}{n}. \tag{5.15}$$

On the inward ray, compactness of the branch and $\Im W(-\epsilon,\pi+\delta)\ge c(\epsilon+\mathrm{dist}(\delta,2\pi\mathbb{Z})^2)$ imply

$$\Re g(-\epsilon,\delta)\ge c(\epsilon+\mathrm{dist}(\delta,2\pi\mathbb{Z})^2). \tag{5.16}$$

Indeed $\Im W=\sinh(\Re g)\sin(\Im g)$, and the relevant roots remain in a fixed compact set. All parameter derivatives of g are bounded on a fixed complex neighborhood.

It follows from (5.15) that C_n is holomorphic on $|t|<c/n^2$. On this disk, $\Re\Gamma_n\ge c/n$ after decreasing c, and hence $|\coth(\Gamma_n/2)|\le Cn$. The torus volume is absorbed in the linear exponential in (5.14). Cauchy's estimate therefore gives

$$|[t^j]C_n(t)|\le(Cn^2)^j\,CnE_n. \tag{5.17}$$

More generally, the disk of radius $c(\epsilon+n^{-2})$ about $-\epsilon$ has $\Re\Gamma_n\ge c(n\epsilon+n^{-1})$. The same argument bounds the jth derivative at $-\epsilon$ by $j!(Cn^2)^jCnE_n$, uniformly for small ϵ. On compact subsets of $\mathcal{U}^-$, the kernel is bounded and (5.14) proves normal convergence. Dominated convergence for each fixed j now proves

$$a_j=\sum_{\substack{n\ge1\\ n\ \mathrm{odd}}}[t^j]C_n(t),\qquad\sum_n|[t^j]C_n(t)|<\infty. \tag{5.18}$$

5.3 An exact reduction to simple-pole periods

Fix a sufficiently small $\sigma > 0$. There is a fixed $c_1 > 0$ such that on $|t| < c_1/n$ the part with $S \geq \sigma$ is analytic with a bounded kernel. To see this, use $\Re\Gamma_n(0,q) \geq cS$ and the perturbation bound $|\Gamma_n(t,q) - \Gamma_n(0,q)| \leq Cn|t|$. Choose c_1 to be a sufficiently small multiple of σ.

In the region $S < \sigma$, set

$$D_n(t,q) = \Gamma_n(t,q) - n\mathrm{i}\phi.$$

The Taylor expansion of g at $(0,0)$ gives $|D_n(t,q)| \leq C(S + n|t|)$. If $p \nmid n$, the distance of $n\phi$ from $2\pi\mathbb{Z}$ is at least $2\pi/p$. Reduce σ, c_1 so that the perturbation is less than π/p. This part is also analytic, with bounded kernel, on $|t| < c_1/n$.

It remains to consider odd multiples of p and $S < \sigma$. Here $n\phi \in 2\pi\mathbb{Z}$, and

$$\coth(\Gamma_n/2) = \frac{2}{D_n} + \mathcal{R}(D_n),$$

where $\mathcal{R}$ is holomorphic on a fixed disk containing all these values for $|t| < c_1/n$. There is a unique small zero $t = T_n(q)$ of $D_n(t,q)$. This statement has a fixed-disk quantitative form. With $a_0 = B(0)/(\mathrm{i}s_\phi) \neq 0$, one has

$$\frac{1}{n}\partial_t D_n(t,q) = a_0 + O(|t| + \sigma)$$

on a fixed disk. By reducing that disk and σ, its distance from a_0 is less than $|a_0|/4$. Rouché's theorem on a circle of radius CS/n, followed by uniqueness from this derivative estimate, gives

$$|T_n(q)| \leq CS/n, \qquad D_n(t,q) = n(t - T_n(q))A_n(t,q), \tag{5.19}$$

where A_n is holomorphic on a fixed smaller disk and bounded away from zero. One can take

$$A_n(t,q) = \frac{1}{n}\int_0^1 \partial_t D_n(T_n(q) + u(t - T_n(q)), q)\ \mathrm{d}u.$$

All such disks and bounds are independent of n and q.

Put $J_n = H_n/A_n$. The exact identity

$$\frac{2H_n(t,q)}{D_n(t,q)} = \frac{2J_n(T_n(q),q)}{n(t - T_n(q))} + \frac{2}{n}\frac{J_n(t,q) - J_n(T_n(q),q)}{t - T_n(q)} \tag{5.20}$$

separates the pole from an analytic divided difference. The second term is bounded by CE_n, by fixed-disk Cauchy estimates. The $\mathcal{R}$ term has the same bound on $|t| < c_1/n$. Combining these observations with the exterior and nonresonant regions gives

$$a_j = \sum_{\substack{n \text{ odd} \\ p|n}} A_{n,j}^{\mathrm{pole}} + R_j, \tag{5.21}$$

$$A_{n,j}^{\mathrm{pole}} = -\int_{S<\sigma} \frac{2H_n(T_n(q),q)}{\partial_t\Gamma_n(T_n(q),q)} T_n(q)^{-j-1}\ \mathrm{d}q, \tag{5.22}$$

$$|R_j| \leq \sum_{n\geq 1} (Cn)^j \exp\{-\kappa n^2 + Cn\log(n+1) + C\} \tag{5.23}$$

$$\leq \exp\{\tfrac{1}{2}j\log(j+2) + Cj + C\}. \tag{5.24}$$

For each n, the coefficient identity is first justified on the smaller disk $|t| < c/n^2$; no pole is crossed there. The last inequality follows by absorbing $Cn\log(n+1)$ into half the Gaussian and maximizing $j\log n - \kappa n^2/2$. This reduction isolates all small pole locations on the original real angular cycle.

5.4 Bounds for every winding and every particle index

The zero in (5.19) has a useful more precise estimate. Set

$$v_i = 1 - \cos\delta_i, \qquad \bar v = \frac{1}{n}\sum_i v_i.$$

At $t = \bar v$, the linear terms in the expansion of $\operatorname{arcosh}(b + B(\bar v)(\bar v - v_i))$ cancel on averaging. The remaining average is bounded by $C\,n^{-1}\sum_i (v_i - \bar v)^2$. The uniform nonzero derivative used in (5.19) therefore yields

$$T_n = \bar v + O\!\left(\frac{1}{n}\sum_i (v_i - \bar v)^2\right).$$

Since $v_i \ge 0$, $\max_i v_i \le C\sigma$, and $\bar v = (S/(2n))(1 + O(\sigma))$, this implies

$$|T_n(q)| \ge (1 - C\sigma)\frac{S}{2n}, \qquad |\partial_t \Gamma_n(T_n(q), q)| \ge cn. \tag{5.25}$$

Retain the exact central numerator size

$$H_{0,n} = \frac{|B(0)|^{n^2/2}}{n!(2\pi)^{n-1}2^{n(n-1)}s_\phi^{n^2}}. \tag{5.26}$$

Dividing the sine factors by their linear terms and all the hyperbolic sine factors by $\mathrm{i}s_\phi$ gives

$$|H_n(T_n(q), q)| \le H_{0,n}\Delta(\delta)^2 e^{CnS}, \qquad \Delta(\delta) = \prod_{i<k}(\delta_i - \delta_k). \tag{5.27}$$

Here is the relevant accounting. The logarithm of the anisotropy ratio costs $Cn^2|T_n| \le CnS$. The sum of the absolute one-body phase deviations is at most CS, so all pair denominators together cost CnS. Finally $\sum_{i<k}|\delta_i - \delta_k|^2 \le CnS$ controls the sine divided differences. Thus no undetermined constant to the power n^2 replaces the displayed normalization.

For an odd winding w, write

$$\delta_i = \frac{w\pi}{n} + \eta_i, \qquad \sum_i \eta_i = 0, \qquad Q(\eta) = \sum_i \eta_i^2, \qquad m_{n,w} = \frac{\pi^2 w^2}{n}. \tag{5.28}$$

Then $S = Q + m_{n,w}$ and $\Delta(\delta) = \Delta(\eta)$. The winding region is exactly $Q < \sigma - m_{n,w}$ when it is nonempty. In particular there are at most $C\sqrt{n}$ windings. Define

$$\alpha_n = \frac{n^2 - 1}{2}, \qquad P_n = \prod_{r=1}^{n-1}\frac{r!}{2^r}, \qquad G_n = \frac{n!\pi^{(n-1)/2}P_n}{\sqrt{n}}. \tag{5.29}$$

On the real zero-sum plane, with measure $\mathrm{d}\eta_1 \cdots \mathrm{d}\eta_{n-1}$,

$$\int e^{-Q(\eta)}\Delta(\eta)^2\,\mathrm{d}\eta = G_n, \tag{5.30}$$

$$\int \frac{\Delta(\eta)^2}{(Q(\eta) + m)^{j+1}}\,\mathrm{d}\eta = G_n m^{\alpha_n - j - 1}\frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)} \quad (j + 1 > \alpha_n). \tag{5.31}$$

To verify the constants, the unrestricted Gaussian Vandermonde integral is $n!\prod_{r=0}^{n-1}(\sqrt{\pi}\,r!/2^r) = n!\pi^{n/2}P_n$. This follows by writing the Vandermonde as a determinant of monic Hermite polynomials

and integrating the determinant product. The monic squared norm is $\sqrt{\pi}\, r!/2^r$, obtained by integration by parts. In the decomposition $x_i = \eta_i + u$, $\sum \eta_i = 0$, the ordinary-coordinate Jacobian is n, and the mean integral is $\int e^{-nu^2}\,\mathrm{d}u = \sqrt{\pi/n}$. This proves (5.30). Homogeneity gives radial density proportional to $Q^{\alpha_n - 1}\,\mathrm{d}Q$, proving (5.31) by the ordinary beta integral.

Let $A^{\mathrm{pole}}_{n,j,w}$ denote the contribution of the wth winding to (5.22). Equations (5.25)–(5.31) imply

$$|A^{\mathrm{pole}}_{n,j,w}| \le \frac{CH_{0,n}}{n} e^{Cn\sigma} \left(\frac{2n}{1 - C\sigma}\right)^{j+1} G_n m_{n,w}^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)} \quad (j + 1 > \alpha_n). \tag{5.32}$$

The dependence on w here is the explicit factor $|w|^{-2(j+1-\alpha_n)}$.

There is also a bound without any condition on j/α_n. Use $|T_n| \ge c/n^2$, and use homogeneity to write

$$\int_{Q<\sigma} \Delta(\eta)^2\ \mathrm{d}\eta = G_n \frac{\sigma^{\alpha_n}}{\Gamma(\alpha_n + 1)}.$$

The elementary factorial bounds in (5.29) then give

$$\log\left(\sum_w |A^{\mathrm{pole}}_{n,j,w}|\right) \le 2j \log n - \frac{1}{2} n^2 \log n + Cn^2 + Cj + C. \tag{5.33}$$

In detail, $\log P_n = \frac{1}{2} n^2 \log n + O(n^2)$ and $\log \Gamma(\alpha_n + 1) = n^2 \log n + O(n^2)$; the remaining factors in $H_{0,n} G_n$ have logarithm $O(n^2)$. The number of windings costs only $O(\log(n+1))$. This proves (5.33) also when n is much larger than the Taylor order.

5.5 Uniform scalar poles on the original real plane

We first make the parameter dependence precise. Let $\mathcal{P}_0 \Subset \mathcal{P}_1$ be concentric parameter polydisks, let $R, M, \beta > 0$, and suppose that

$$B \in \mathcal{O}(\{|t| < R\} \times \mathcal{P}_1), \qquad |B(t,\eta)| \le M, \qquad \Im B(0,\eta) \le -\beta \quad (\eta \in \mathcal{P}_1).$$

For a relatively compact admissible parameter set the sign hypothesis can always be arranged on a slightly larger polydisk by continuity; a finite cover suffices if several charts are needed. Every constant below is independent of $\eta \in \mathcal{P}_0$, n, and j, unless another dependence is displayed. Constants may depend on the fixed phase, R, M, β, and the polydisk collar.

Lemma 5.2 (Common scalar neighborhoods)**.** *There are common positive parameter and angular radii on which the branch $g(t,\delta,\eta)$, all fixed scalar derivatives needed below, the nonzero one-body and canceled-pair denominators, and the logarithm of B are uniformly controlled. The estimates* (5.14)*,* (5.17)*,* (5.24)*,* (5.32)*, and* (5.33) *consequently hold with common constants. For each fixed j and parameter multiindex ν,*

$$\sum_{n\ \mathrm{odd}} \sup_{\eta \in \mathcal{P}_0} \left|\partial_\eta^\nu [t^j] C_n(t,\eta)\right| < \infty.$$

The same majorant permits the inward boundary limit for these fixed mixed derivatives.

Proof. On $|t| \le R/2$, Cauchy's inequality bounds B_t by $2M/R$. Choose $r > 0$ so small that $|B(t,\eta) - B(0,\eta)| \le \beta/2$ for $|t| < r$. Thus $-\Im B \ge \beta/2$ and $|B| \ge \beta/2$ there. The lower half-plane supplies a common logarithm, whose possible physical sign change multiplies every odd coefficient by the same nonzero sign.

The compact set

$$\{b - Bv : |B| \le M,\ \Im B \le -\beta,\ 0 \le v \le 2\}$$

avoids both branch cuts of the branch of arccos through ϕ: at $v = 0$ it equals $b \in (-1, 1)$, and at $v > 0$ its imaginary part is positive. It therefore has a fixed neighborhood on which $g = \mathrm{i} \arccos W$ is holomorphic. Uniform continuity of cosine gives common smaller t and complex-angle neighborhoods. At $t = 0$ the corresponding roots form a compact subset of the closed unit disk strictly inside its lower half-plane. The strict root-Schur estimate follows uniformly from (5.13), and persists after one further common shrink. All canceled factors and one-body denominators have uniform bounds on this neighborhood. Cauchy's inequality supplies every fixed scalar or parameter jet.

The proofs of the cited estimates use just these bounds, a fixed lower bound for $|B(0)/\sin\phi|$, and the fixed rational phase. Thus their disk sizes, σ, Gaussian exponent, and constants can be chosen in common. For parameter derivatives first work on an intermediate polydisk $\mathcal{P}_0 \Subset \mathcal{P}_* \Subset \mathcal{P}_1$. Cauchy's inequality in its fixed collar bounds each $\partial_\eta^\nu [t^j] C_n$ by a constant times $(Cn^2)^j C n E_n$. This is summable for fixed j, ν, including on the disks of radius $c(\epsilon + n^{-2})$ about $-\epsilon$. Dominated convergence proves the assertions. □

For an odd multiple n of p, put

$$d_n = \frac{\pi}{n}, \qquad \tau_n = 1 - \cos d_n, \qquad B_n = B(\tau_n, \eta), \qquad h_n = \cos d_n - \frac{B_n b}{s_\phi^2} \sin^2 d_n, \qquad a_n = \Re h_n. \tag{5.34}$$

Uniformly, $h_n = 1 + O(n^{-2})$ and $a_n > 0$ for sufficiently large n. In the first winding write

$$\delta_i = d_n + v_i, \qquad V_n = \{v \in \mathbb{C}^n : \sum_i v_i = 0\}, \qquad Q(v) = \sum_i v_i^2.$$

On the real plane the measure is $\mathrm{d}v_1 \cdots \mathrm{d}v_{n-1}$, with $v_n = -\sum_{i<n} v_i$. The exact first-winding region $S < \sigma$ is

$$Q < r_n^2, \qquad r_n^2 = \sigma - \frac{\pi^2}{n}. \tag{5.35}$$

For large n, its squared radius lies between $\sigma/2$ and σ.

The exact scalar pole is determined by

$$\sum_i \big(g(T_n(v), d_n + v_i, \eta) - \mathrm{i}\phi\big) = 0.$$

Subtracting the common constant and linear terms in the v_i bounds the remaining sum by $C \sum_i |v_i|^2$. The parameter derivative of its average stays in a fixed disk around $g_t(\tau_n, d_n, \eta) = -\mathrm{i}B_n/s_\phi$, bounded away from zero. Rouché's theorem on the circle $|t - \tau_n| = C \sum_i |v_i|^2/n$, followed by uniqueness, gives a holomorphic solution on a common small complex Euclidean ball in V_n, jointly with η, and

$$T_n(0) = \tau_n, \qquad |T_n(v) - \tau_n| \le \frac{C}{n} \sum_i |v_i|^2. \tag{5.36}$$

The scalar derivative bounds are taken on fixed neighborhoods independent of the dimension.

At the center direct differentiation gives

$$g(\tau_n, d_n) = \mathrm{i}\phi, \qquad g_t(\tau_n, d_n) = -\frac{\mathrm{i}B_n}{s_\phi}, \qquad \frac{1}{2} g_{\delta\delta}(\tau_n, d_n) = \frac{\mathrm{i}B_n h_n}{2 s_\phi}.$$

In $W_t = B(t) + B_t(t)(t - 1 + \cos\delta)$, the second term vanishes exactly at this center; no derivative of B was frozen. Evenness in δ gives $g_{\delta\delta\delta}(t, d_n) = O(d_n)$ uniformly. Writing $u = T_n - \tau_n$, Taylor's formula and $\sum_i v_i = 0$ yield

$$0 = ng_t(\tau_n, d_n)u + \tfrac{1}{2}g_{\delta\delta}(\tau_n, d_n)Q + O\left(n|u|^2 + |u|Q + d_n\sum_i |v_i|^3 + \sum_i v_i^4\right)$$

on the real ball. Inserting (5.36) proves

$$\begin{aligned} T_n(v) &= T_{0,n}(v) + \mathcal{R}_n(v), \qquad T_{0,n}(v) = \tau_n + \frac{h_n Q}{2n}, \\ |\mathcal{R}_n(v)| &\le \frac{C}{n}\left(\frac{1}{n}\sum_i |v_i|^3 + \sum_i v_i^4 + \frac{Q^2}{n}\right). \end{aligned} \tag{5.37}$$

This estimate retains the fourth-power sum: replacing it pointwise by CQ^2/n would be false. Since $\sum |v_i|^3 \le Q^{3/2}$ and $\sum v_i^4 \le Q^2$, a common sufficiently small σ and large n give

$$|\mathcal{R}_n(v)| \le \tfrac{1}{4}D_n(v), \qquad D_n(v) = \tau_n + \frac{a_n Q}{2n} \le |T_{0,n}(v)|, \qquad Q < \sigma. \tag{5.38}$$

5.6 The complete residue and its angular measure

In this first winding put

$$\varrho_n(v) = -\frac{2H_n(T_n(v), \pi + d_n + v)}{\partial_t\Gamma_n(T_n(v), \pi + d_n + v)}.$$

All factors in H_n, including $B(T_n, \eta)^{n^2/2}$, are retained. After filling the sine divided differences, $\varrho_n/\Delta(v)^2$ is holomorphic and nonzero on a common small ball. Its exact center value is

$$\varrho_{n,0} = -\frac{2B_n^{\alpha_n - 1/2}}{n\, n!(2\pi)^{n-1}2^{n(n-1)}s_\phi^{2\alpha_n}}, \qquad \alpha_n = \frac{n^2 - 1}{2}. \tag{5.39}$$

Indeed $\mathrm{i}^{-n^2} = -\mathrm{i}$ for odd n, so the centered numerator divided by Δ^2 is

$$-\frac{\mathrm{i}B_n^{n^2/2}}{n!(2\pi)^{n-1}2^{n(n-1)}s_\phi^{n^2}}, \qquad \partial_t\Gamma_n = -\frac{\mathrm{i}nB_n}{s_\phi}.$$

The remaining minus sign comes from the Taylor expansion of $(t - T_n)^{-1}$.

Lemma 5.3 (A direct residue-density bound)**.** *On the real ball $Q < \sigma$,*

$$\varrho_n(v) = \varrho_{n,0}\Delta(v)^2 e^{\ell_n(v)}, \qquad |\ell_n(v)| \le CnQ. \tag{5.40}$$

The logarithm has a holomorphic extension through the coincidence divisors, and the estimate is uniform in the parameter family.

Proof. Let $e_i = g(T_n, d_n + v_i) - \mathrm{i}\phi$. The exact pole equation gives $\sum_i e_i = 0$, and the scalar Taylor bounds give

$$|e_i| \le C(|u| + |d_n v_i| + |v_i|^2), \qquad \sum_i |e_i|^2 \le C\left(\frac{Q^2}{n} + \frac{Q}{n^2} + \sum_i v_i^4\right).$$

On a common small disk the function $L(z) = \log(\sinh(\mathrm{i}\phi + z)/(\mathrm{i}s_\phi))$ is holomorphic and $L(z) = L'(0)z + O(|z|^2)$. Thus the linear parts in both $\sum_i L(e_i)$ and $2\sum_{i<k} L((e_i + e_k)/2)$ vanish exactly. Their total absolute logarithmic cost is at most $Cn\sum_i |e_i|^2 \le CnQ$ on the fixed ball. The sine divided differences cost $C\sum_{i<k}(v_i - v_k)^2 = CnQ$. The anisotropy factor costs $Cn^2|u| \le CnQ$. Finally, summing the scalar Taylor expansion of $g_t(T_n, d_n + v_i)$ cancels its linear angular term and gives

$$\frac{\sum_i g_t(T_n, d_n + v_i)}{n g_t(\tau_n, d_n)} = 1 + O(Q/n).$$

Taking the logarithm of each already normalized nonzero factor proves the result. In particular no Jacobian or nonlinear angular change of variables enters this density. □

Separate radius and angle in the positive real measure $\Delta(v)^2\,\mathrm{d}v$: write $v = \sqrt{Q}\,\theta$, with $\sum_i \theta_i = 0$ and $\sum_i \theta_i^2 = 1$. Let ν_n be the normalized angular measure proportional to $\Delta(\theta)^2$.

Lemma 5.4 (A scalar angular exponential moment)**.** *For $Z_n = n\max_i \theta_i^2$, there are absolute constants $c_0, C_0 > 0$ such that*

$$\sup_{n\ge 3} \mathbb{E}_{\nu_n}[(1 + Z_n)e^{c_0 Z_n}] \le C_0. \tag{5.41}$$

One may take $c_0 = 1/32$.

Proof. The constrained Gaussian probability measure $G_n^{-1}e^{-Q}\Delta(v)^2\,\mathrm{d}v$ has the normalization (5.29). Homogeneity shows that Q and θ are independent, the angular law is ν_n, and Q has gamma density $q^{\alpha_n - 1}e^{-q}/\Gamma(\alpha_n)$.

Fix $v_1 = x$, let $m = n - 1$ and $c = n/(n-1)$, and write $v_{i+1} = u_i - x/m$, $\sum_i u_i = 0$. The independent-coordinate Jacobian is one, and

$$Q(v) = cx^2 + Q(u), \qquad \Delta(v)^2 = \Delta(u)^2 \prod_{i=1}^{m}(cx - u_i)^2.$$

Hence the density of the first coordinate is

$$f_n(x) = \frac{G_{n-1}}{G_n} e^{-cx^2}\mathbb{E}_m \prod_{i=1}^{m}(cx - u_i)^2, \qquad \frac{G_{n-1}}{G_n} = \frac{2^{n-1}}{\sqrt{\pi}\,n!}\sqrt{\frac{n}{n-1}}.$$

By arithmetic–geometric mean and $\sum_i u_i = 0$,

$$\prod_i (cx - u_i)^2 \le \left(c^2x^2 + \frac{Q(u)}{m}\right)^m.$$

For $0 \le k \le m$, the gamma moments obey

$$\mathbb{E}_m\left(\frac{Q(u)}{m}\right)^k = \frac{\alpha_m(\alpha_m + 1)\cdots(\alpha_m + k - 1)}{m^k} \le n^k,$$

since $(\alpha_m + m - 1)/m \le n$. Using $c^2 < 3$, the binomial expansion therefore bounds the expected product by $(3x^2 + n)^{n-1}$. Since $n! \ge (n/e)^n$, with $t = x^2/n$,

$$f_n(x) \le \frac{1}{n}(2e)^n(1 + 3t)^{n-1}e^{-x^2}.$$

For $t \ge 16$, $\log(2e) + \log(1+3t) \le t/2$; the inequality holds at 16, and the difference has positive derivative thereafter. It follows that $f_n(x) \le n^{-1}e^{-x^2/2}$ when $x^2 \ge 16n$. Permutation invariance of the constrained measure and the union bound now give

$$\Pr\{\max_i v_i^2 \ge A^2\} \le 2\int_A^\infty e^{-x^2/2}\,\mathrm{d}x \le \frac{2}{A}e^{-A^2/2}, \qquad A^2 \ge 16n.$$

On the other hand,

$$\Pr\{Q < n^2/8\} \le \frac{n^2}{8}\mathbb{E}(Q^{-1}) = \frac{n^2}{4(n^2-3)} \le \frac{3}{8} \qquad (n \ge 3).$$

Independence of radius and angle and $\max_i v_i^2 = QZ_n/n$ consequently imply

$$\frac{5}{8}\Pr\{Z_n \ge z\} \le \Pr\{\max_i v_i^2 \ge nz/8\}.$$

For $z \ge 128$, the preceding Gaussian tail gives $\Pr\{Z_n \ge z\} \le (16/5)e^{-nz/16}$. Integrating this tail, and bounding $Z_n \le 128$ directly on the complementary event, proves (5.41). □

The consequences used below are the pointwise inequalities

$$\sum_i v_i^4 \le \frac{Q^2 Z_n}{n}, \qquad \sum_i |v_i|^3 \le \sqrt{Q\sum_i v_i^4} \le \frac{Q^{3/2}\sqrt{Z_n}}{\sqrt{n}}. \tag{5.42}$$

The exponential moment includes the rare directions where Z_n is large.

5.7 An exact complex beta period and the two first windings

For $j+1 > \alpha_n$ and $\Re h > 0$, the same constrained Gaussian period and the scalar Laplace formula give

$$J_h := \int_{V_n \cap \mathbb{R}^n} \frac{\Delta(v)^2\,\mathrm{d}v}{(\tau_n + hQ/(2n))^{j+1}} = G_n(2n)^{\alpha_n} h^{-\alpha_n} \tau_n^{\alpha_n - j - 1}\frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)}. \tag{5.43}$$

Indeed the radial measure is $G_n Q^{\alpha_n - 1}\,\mathrm{d}Q/\Gamma(\alpha_n)$. Applying the Laplace representation of the negative power and then integrating this radial gamma integral proves the formula. Absolute convergence follows from $\Re h > 0$ and $j + 1 > \alpha_n$, so every integration is on the original real variables. In particular the gamma quotient is exact.

For the positive majorant use $a_n = \Re h_n$. The exact periods satisfy

$$\frac{J_{a_n}}{|J_{h_n}|} = \left(\frac{|h_n|}{a_n}\right)^{\alpha_n} = \exp(O(n^{-2})). \tag{5.44}$$

This is uniform: $a_n = 1 + O(n^{-2})$, $\Im h_n = O(n^{-2})$, and therefore $\log(|h_n|/a_n) = O(n^{-4})$. The comparison has no factor growing with j.

We shall need a relative estimate only in the entire intermediate range

$$\log j < n < \frac{\sqrt{j}}{(\log j)^{1/4}}. \tag{5.45}$$

Here both $n \to \infty$ and $\epsilon_n = n^2/j \to 0$ hold uniformly. The exact first-winding contribution is

$$I_{n,j} = \int_{Q < r_n^2} \varrho_n(v) T_n(v)^{-j-1}\,\mathrm{d}v.$$

We prove

$$I_{n,j} = \varrho_{n,0} J_{h_n} \left(1 + O\left(\frac{n}{\sqrt{j}}\right)\right) \tag{5.46}$$

uniformly in (5.45) and the parameter family.

Normalize the positive measure for J_{a_n} and set

$$x = \frac{a_n Q}{2n\tau_n}, \qquad \epsilon = \frac{n^2}{j}.$$

Radius and angle remain independent. The angular law is ν_n, and the radial density is proportional to $x^{\alpha_n - 1}(1+x)^{-j-1}$. Since $2n^2\tau_n$ and a_n are bounded above and below, $Q \asymp x/n$. Equations (5.37) and (5.42) give

$$(j+1)\frac{|\mathcal{R}_n|}{D_n} \le C\frac{j}{n^2}\frac{x^{3/2}\sqrt{Z_n} + x^2(Z_n+1)}{1+x}. \tag{5.47}$$

Because $|\mathcal{R}_n/T_{0,n}| \le 1/4$, its near-one logarithm has modulus at most $2|\mathcal{R}_n|/D_n$. Thus

$$\left|\frac{e^{\ell_n}}{T_n^{j+1}} - \frac{1}{T_{0,n}^{j+1}}\right| \le D_n^{-j-1}\left[\exp\left\{|\ell_n| + \frac{2(j+1)|\mathcal{R}_n|}{D_n}\right\} - 1\right]. \tag{5.48}$$

Fix $K = 32$. On $x \le K\epsilon$, (5.40) gives $|\ell_n| \le Cx \le C\epsilon$, and (5.47) bounds the exponent by

$$C\{\sqrt{\epsilon}\sqrt{Z_n} + \epsilon(Z_n+1)\} \le C'\sqrt{\epsilon}(1+Z_n).$$

For uniformly large j, its coefficient is less than c_0. The mean-value theorem for the exponential and Lemma 5.4 therefore show that the angular expectation of the bracket in (5.48) is $O(\sqrt{\epsilon})$. This controls all angular directions in the central radial region.

On the actual ball, $x \le Ln$ for a fixed uniform L. The deterministic bound $Z_n \le n$ gives

$$\frac{2(j+1)|\mathcal{R}_n|}{D_n} \le Cj\frac{x^{3/2}/n^{3/2} + x^2/n}{1+x}.$$

Dividing by $j\log(1+x)$, the right side tends to zero uniformly on $0 < x \le Ln$: the intervals $(0,1]$, $[1,\sqrt{n}]$, and $[\sqrt{n}, Ln]$ give respectively $O(n^{-1})$, $O(n^{-1/2})$, and $O(1/\log n)$. Also $Cx/[j\log(1+x)] = o(1)$, using $j \to \infty$ below one and $n/j \to 0$ above one. Hence on the remaining radial region

$$|\ell_n| + \frac{2(j+1)|\mathcal{R}_n|}{D_n} \le \frac{j}{4}\log(1+x), \qquad K\epsilon \le x \le Ln.$$

Write $Z_\beta(\alpha, J) = \Gamma(\alpha)\Gamma(J+1-\alpha)/\Gamma(J+1)$. Dividing the resulting tail integral by the original normalization bounds it by

$$C(1+K\epsilon)^{-j/4}\frac{Z_\beta(\alpha_n, j/2)}{Z_\beta(\alpha_n, j)} \le C(1+K\epsilon)^{-j/4}4^{\alpha_n} \le Ce^{-cn^2}.$$

For the middle inequality, α_n is an integer and $\alpha_n \le j/4$ eventually, so the exact ratio is

$$\prod_{r=0}^{\alpha_n - 1}\frac{j-r}{j/2-r} \le 4^{\alpha_n}.$$

For the last inequality use $\log(1+K\epsilon) \ge K\epsilon/2$. The omitted part of the unperturbed period outside $Q < r_n^2$ has the same bound: its radial lower endpoint is at least a positive constant times n, hence exceeds $K\epsilon$, and its positive majorant has no perturbation factor.

Combining these estimates with (5.44) gives relative error at most $C(n/\sqrt{j} + e^{-cn^2})$. The lower bound $n > \log j$ absorbs the second term into the first. This proves (5.46) on the exact original ball; no cycle-boundary remainder was introduced.

Reflection of all lifts $\delta_i \mapsto -\delta_i$ preserves the dispersion, every squared sine factor, and the independent real measure. It exchanges the first two windings. Their integrals are therefore equal, with their full complex phase. Substitution of (5.39) and (5.29) into (5.43) gives

$$A^{\mathrm{pole}}_{n,j,1} + A^{\mathrm{pole}}_{n,j,-1} = M(n,j)(1 + e_{n,j}), \tag{5.49}$$

$$|e_{n,j}| \le C\frac{n}{\sqrt{j}}, \tag{5.50}$$

$$M(n,j) = -\frac{2^{2-\alpha_n} B_n^{\alpha_n - 1/2} P_n n^{\alpha_n - 3/2}}{\pi^{(n-1)/2} s_\phi^{2\alpha_n}} h_n^{-\alpha_n} \tau_n^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)}. \tag{5.51}$$

This is the same exact-factor model for the whole residue, including $B(\tau_n, \eta)$ and its particle-dependent complex phase. There is no positivity assumption across different particle indices.

5.8 Integer selection of a single dominant particle index

Let $L(n,j) = \log|M(n,j)|$ and $d = 2p$. All estimates in this subsection are uniform in the parameter family fixed above; the selected integer may depend on η. We use only integer Taylor orders and retain the exact gamma quotient.

For $A = \pi^2/2$, put

$$e_n = \log(\tau_n n^2/A), \qquad E_j(a) = \sum_{r=0}^{a-1} -\log(1 - r/j).$$

For $j > \alpha_n$, direct grouping of the factors in (5.51) gives

$$L(n,j) = \log P_n + (2j - n^2/2 + 1)\log n - \alpha_n \log j - j\log A + R_n - je_n + E_j(\alpha_n). \tag{5.52}$$

Here R_n is a quadratic polynomial in n, plus a convergent power series in n^{-2} of order $O(n^{-2})$. Its coefficients are uniformly bounded. Indeed $\tau_n n^2/A$, $B(\tau_n,\eta)/B(0,\eta)$, and h_n have common convergent expansions in n^{-2} near zero, with uniformly nonzero constant terms. Their near-one logarithms have common convergent expansions; multiplication by α_n introduces at most a constant term. In particular $e_n = O(n^{-2})$, its centered first and second differences are $O_d(n^{-3})$ and $O_d(n^{-4})$, and those of R_n are $O_d(n)$ and $O_d(1)$.

Ordinary Stirling bounds and $E_j(\alpha_n) = O(n^4/j)$ give, as $n \to \infty$ and $n^2/j \to 0$,

$$L(n,j) = 2j\log n - \frac{n^2}{2}\log j - j\log(\pi^2/2) + O(n^2 + n^4/j + j/n^2). \tag{5.53}$$

The finite differences for the comparison are computed from (5.52).

$$L(n+d,j) - 2L(n,j) + L(n-d,j) = -d^2\left(\frac{2j}{n^2} + \log j\right) + O_d(1 + n^2/j + j/n^4), \tag{5.54}$$

$$\frac{L(n+d,j) - L(n-d,j)}{2d} = \frac{2j}{n} - n\log j + O_d(n + n^3/j + j/n^3). \tag{5.55}$$

These hold for $j > \alpha_{n+d}$ in the stated regime.

The factorial products have the exact differences

$$\log P_{n+d} - \log P_{n-d} = \sum_{r=n-d}^{n+d-1} \big(\log(r!) - r\log 2\big),$$

$$\log P_{n+d} - 2\log P_n + \log P_{n-d} = \sum_{k=0}^{d-1}\sum_{\ell=1}^{d} \log(n-d+k+\ell) - d^2\log 2.$$

Stirling's inequality on the finitely many factorials makes the first $2dn\log n + O_d(n)$; the second is $d^2\log n + O_d(1)$. The corresponding elementary differences of $(2j - n^2/2 + 1)\log n$ cancel these $\log n$ terms, leaving $2j/n + O_d(n + j/n^3)$ in the centered first difference divided by $2d$, and $-2d^2 j/n^2 + O_d(1 + j/n^4)$ in the second.

For the gamma correction, on $0 \le r \le \alpha_{n+d}$,

$$0 \le -\log(1 - r/j) \le Cn^2/j, \qquad \frac{\mathrm{d}}{\mathrm{d}r}[-\log(1-r/j)] \le C/j.$$

The difference $E_j(\alpha_{n+d}) - E_j(\alpha_{n-d})$ has $2dn$ summands and is $O_d(n^3/j)$. For the second difference, the upper and lower increments of α_n have lengths $dn + d^2/2$ and $dn - d^2/2$. Their d^2 unmatched summands cost $O_d(n^2/j)$; pairing the others, whose arguments differ by $O_d(n)$, gives the same bound. Finally the differences of $-\alpha_n \log j$ are exactly $-n\log j$ after division by $2d$, and $-d^2\log j$. The bounds on R_n and je_n now prove (5.54)– (5.55).

Throughout (5.45), the second difference is at most $-cd^2\log j$ for all sufficiently large j, whenever the neighboring indices lie in that range. Indeed $n^2/j = o(1)$ uniformly, whereas j/n^4 is smaller than j/n^2 by the factor $n^{-2} \le (\log j)^{-2}$.

Fix a large allowed n, and for integer $j > \alpha_{n+d}$ set $F_n(j) = L(n+d, j) - L(n-d, j)$. The one-step gamma recurrence gives the exact identity

$$F_n(j+1) - F_n(j) = \log\frac{\tau_{n-d}}{\tau_{n+d}} + \log\frac{j+1-\alpha_{n+d}}{j+1-\alpha_{n-d}}. \tag{5.56}$$

On the integer interval $\frac{1}{2}n^2\log n \le j \le \frac{3}{2}n^2\log n$, this is $4d/n + O_d(1/(n\log n))$, hence positive and $O_d(1/n)$. It is independent of η.

At the two endpoints $j = cn^2\log n + O(1)$, $c = 1/2, 3/2$, the leading part of (5.55) is

$$\frac{2j}{n} - n\log j = 2(c-1)n\log n - n\log\log n - n\log c + O(1/n).$$

The error there is $O_d(n)$, uniformly in η, so these endpoint signs are opposite. Define $j_n(\eta)$ as the first integer in this interval for which $F_n(j_n) \ge 0$. Then (5.56) gives

$$j_n \sim n^2\log n, \qquad 0 \le F_n(j_n) = O_d(1/n), \tag{5.57}$$

uniformly in the family. For the first conclusion insert the second in (5.55) and use $\log j_n = 2\log n + O(\log\log n)$. These n and their neighbors lie strictly inside (5.45).

Let $\lambda_k = L(n + kd, j_n)$ on the admissible indices in that range. Their successive differences decrease by at least $cd^2\log j_n$, and $\lambda_1 - \lambda_{-1} = O_d(1/n)$. Therefore

$$\lambda_0 - \lambda_{\pm 1} \ge c' d^2\log j_n, \qquad \lambda_k - \lambda_0 \le -c'' d^2 k^2\log j_n + O_d(|k|/n) \quad (k \ne 0).$$

Summing this discrete Gaussian gives

$$\sum_{\substack{m\neq n,\ m\text{ odd},\ p\mid m\\ m\text{ satisfies (5.45)}}} |M(m,j_n)| \le |M(n,j_n)|\, O(j_n^{-c'''d^2}). \tag{5.58}$$

The modulus estimate allows arbitrary phases. The integer selection is made at fixed η, with the bounds uniform on the parameter polydisk.

5.9 Exhausting the remaining indices and windings

We compare every part of (5.18) with the selected model. First, (5.53) and (5.57) imply

$$\log|M(n,j_n)| = j_n\log j_n - j_n\log\log j_n + j_n\big(\log(4/\pi^2)-1\big) + o(j_n), \tag{5.59}$$

$$|M(n,j_n)|^{1/j_n} \sim \frac{4j_n}{e\pi^2\log j_n}. \tag{5.60}$$

Indeed $j_n \sim n^2\log n$ gives

$$2\log n = \log j_n - \log\log j_n + \log 2 + o(1), \qquad \frac{n^2\log j_n}{2j_n} \longrightarrow 1.$$

Substituting these identities in (5.53) gives (5.59); every error there is $o(j_n)$.

The following estimates cover all the complementary terms.

Small particle indices. For $m \le \log j$, the full coefficient bound (5.17) gives

$$\sum_{\substack{m\le\log j\\ m\text{ odd}}} |[t^j]C_m(t)| \le \exp\{2j\log\log(j+2) + Cj + C\}. \tag{5.61}$$

This is negligible compared with (5.59) when $j = j_n$. Equivalently, after the pole reduction one can apply the same bound to the poles plus their analytic remainders, since their difference has the smaller bound (5.24).

All analytic remainders. The sum of the regular remainders in (5.24) is bounded by

$$\exp\{\tfrac{1}{2}j\log(j+2) + Cj + C\}. \tag{5.62}$$

This too is negligible at the selected orders. This estimate includes the entire nonresonant particle class $p\nmid m$ outside the small-index range.

All sufficiently large particle indices. For

$$m \ge \frac{\sqrt{j}}{(\log j)^{1/4}},$$

use (5.33), which has no restriction on m^2/j. With $x = m^2$, its exponent is at most

$$j\log x - \frac{1}{4}x\log x + Cx + Cj + C.$$

Its derivative in x is negative on $x \geq j/\sqrt{\log j}$, for all large j, and at the lower endpoint its value is

$$j \log j - \frac{1}{2} j \log\log j - \frac{1}{4} j\sqrt{\log j} + o(j\sqrt{\log j}). \tag{5.63}$$

For example its derivative is at most $-c \log j$ there for large j, since $j/x \leq \sqrt{\log j}$ and $\frac{1}{4}\log x \sim \frac{1}{4}\log j$. Summing the resulting decreasing exponential over the discrete values $x = m^2$ costs at most a harmless polynomial factor. Thus (5.63) bounds the logarithm of the whole infinite large-index tail up to $o(j\sqrt{\log j})$. Its gap below (5.59) tends to infinity faster than j.

Higher odd windings in the intermediate range. Let $M_0(m, j) > 0$ be the modulus of the expression (5.51) with B_m replaced by $B(0)$, h_m by 1, and τ_m by $\pi^2/(2m^2)$. The exact factors give

$$\left| \log \frac{M_0(m, j)}{|M(m, j)|} \right| \leq C(1 + j/m^2). \tag{5.64}$$

For the τ factor this follows from $\log(\tau_m/(\pi^2/(2m^2))) = O(m^{-2})$; the B and h factors cost $O(1)$. Comparison of (5.32) with this explicit M_0 yields, for each odd $|w| \geq 3$,

$$|A^{\text{pole}}_{m,j,w}| \leq C M_0(m, j) e^{Cm\sigma} (1 - C\sigma)^{-j-1} |w|^{-2(j+1-\alpha_m)}.$$

The constants C here do not depend on m, j, w. Choose the fixed σ still smaller, if necessary, so that $-\log(1 - C\sigma) < \frac{1}{2}\log 3$. Throughout (5.45),

$$\alpha_m = o(j), \qquad m = o(j), \qquad j/m^2 = o(j)$$

uniformly. The number of possible windings is at most $C\sqrt{m}$. Hence for some $c > 0$,

$$\sum_{\substack{w \text{ odd} \\ |w| \geq 3}} |A^{\text{pole}}_{m,j,w}| \leq |M(m, j)| e^{-cj} \tag{5.65}$$

uniformly in that range. The choice of a smaller σ merely changes the fixed constants in the earlier arguments.

The first windings of every intermediate index. Equation (5.50) tends uniformly to zero in (5.45); it is at most $C(\log j)^{-1/4}$. At $j = j_n$, (5.58) shows that the sum of the moduli of all intermediate models other than $M(n, j_n)$ is $o(|M(n, j_n)|)$. The sum of all their relative errors is bounded by

$$o(1) \sum_m |M(m, j_n)| = o(|M(n, j_n)|).$$

The same is true of all higher windings by (5.65).

Substitution of the estimates for all odd indices, windings and analytic remainders into (5.18) gives

$$a_{j_n} = M(n, j_n)(1 + o(1)). \tag{5.66}$$

5.10 Nonextension and uniformity

Equations (5.60) and (5.66) prove (5.11), so $\limsup_j |a_j|^{1/j} = \infty$. If F admitted a holomorphic extension to a disk about zero from the physical side $\mathcal{U}^-$, its derivatives on the inward negative ray would agree with those of that extension. By (5.9), its Taylor coefficients at zero would therefore be the a_j. Cauchy's estimate on any smaller extension disk would bound $|a_j| \le CR^{-j}$, a contradiction. If $P(0) \neq 0$, a holomorphic extension of PF would, after division by the nonvanishing holomorphic factor P, give an extension of F. Thus a local physical normalization by such a factor cannot remove the obstruction.

Finally, Lemma 5.2 supplies common scalar neighborhoods and fixed mixed-derivative majorants for bounded holomorphic families on common polydisks. The real pole, residue and beta estimates use these constants and the absolute angular moment on a fixed contour. The exact finite differences have convergent expansions uniform on the same polydisk collar. The sign bracket and integer recurrence give uniform selection and comparison with $B(\tau_n, \eta)$ throughout and with the integer $j_n(\eta)$ chosen at each parameter. For a nonramified transversal use its holomorphic $B(t)$. On a compact admissible family of curve coordinates the local inverse is defined on common smaller polydisks. This completes the proof of Theorem 5.1.

6 The isotropic restriction on the complete constrained cycle

Here $\lambda = 1$, $s = \sinh z$, and the dispersion parameter will be written E to distinguish it from the notation of Section 3. The full product specializes to $B = 1$ and $C = s + s^{-1}$.

For $E \in \mathbb{C} \setminus [-2, 2]$ and $y = e^{\mathrm{i}\theta}$, let

$$W = E - \cos\theta, \qquad \rho + \rho^{-1} = 2W, \quad |\rho| < 1, \qquad \sinh\gamma = (\rho^{-1} - \rho)/2.$$

The root is unique and holomorphic: a root on the unit circle would force $W \in [-1, 1]$, and the two roots are reciprocal. Write

$$P_{ij} = -\frac{(y_i - y_j)^2 \rho_i \rho_j}{y_i y_j (1 - \rho_i \rho_j)^2}, \qquad Z = \prod_i \rho_i, \qquad a_+ = 0, \quad a_- = \pi.$$

For either sign $\sigma \in \{1, -1\}$ define the exact auxiliary coefficient

$$\mathcal{F}_N^\sigma(E) = \frac{1}{N!} \int_{\mathbb{H}_N^\sigma} \frac{1 + \sigma Z}{1 - \sigma Z} \prod_i \frac{1}{\sinh\gamma_i} \prod_{i<j} P_{ij}\, d\mu_N, \qquad \mathbb{H}_N^\sigma = \left\{ \sum_i \theta_i = a_\sigma \pmod{2\pi} \right\}, \tag{6.1}$$

where

$$d\mu_N = \prod_{i<N} \frac{d\theta_i}{2\pi}, \qquad \theta_N = a_\sigma - \sum_{i<N} \theta_i \pmod{2\pi}.$$

Thus this is Euclidean hypersurface measure divided by $\sqrt{N}(2\pi)^{N-1}$. For $N = 1$ the constrained integral is evaluation at its single angle. All pair factors are rational squares, so no choice of half-angle or logarithm changes the coefficient. Put

$$C_N^\sigma(t) = \mathcal{F}_N^\sigma(t + t^{-1}), \qquad |t| > 1. \tag{6.2}$$

Write $E(\zeta) = \zeta + \zeta^{-1}$ and put $C_N = C_N^+$. The compact normal-convergence bounds for the full product apply to each fixed differentiated series here. To see directly that the auxiliary sign causes

no change, on a compact dispersion neighborhood all roots satisfy $|\rho_i| \le q < 1$ and have imaginary parts of one sign. The latter fact follows from $\Im W_i = \Im\rho_i(1-|\rho_i|^{-2})/2$. The identity

$$|1-vw|^2 - |v-w|^2 = (1-|v|^2)(1-|w|^2) + 4\Im v\Im w$$

therefore bounds every root Schur factor $|(\rho_i-\rho_j)/(1-\rho_i\rho_j)|$ strictly below one, uniformly. Expand the angular Schur product by the matching identity proved in Section 3 and cancel each matched angular pole with its corresponding root factor. The resulting complete matched pairs are bounded; all but $O(N)$ root factors retain the strict contraction. The number of matchings costs $\exp(O(N\log(N+1)))$, while the unmatched root factors supply $\exp(-cN^2)$. This gives a summable coefficient bound on the normalized constrained torus for either sign. All one-body factors and $1-\sigma Z$ have fixed gaps there. Cauchy's inequality on a slightly larger compact neighborhood gives every fixed derivative.

Angle reflection preserves $\mathbb{H}_N^\sigma$, its Haar measure and each complete factor in (6.1). Conjugation also commutes with the coefficients. Replacing E, y_i, ρ_i by $-E, -y_i, -\rho_i$ gives the precise negation identity

$$\mathcal{F}_N^\sigma(-E) = (-1)^N \mathcal{F}_N^{(-1)^N\sigma}(E). \tag{6.3}$$

Every pair is unchanged, every one-body factor changes sign, and $\prod_i y_i$ and Z both acquire $(-1)^N$. This proves (6.3) with its constraint and product-kernel signs. In particular the even physical coefficients are even in t, whereas the odd coefficients satisfy

$$C_N^+(-t) = -C_N^-(t) \qquad (N \text{ odd}). \tag{6.4}$$

The two one-particle formulas, useful below, are

$$\mathcal{F}_1^+(E) = \frac{1}{E-2}, \qquad \mathcal{F}_1^-(E) = \frac{1}{E+2}. \tag{6.5}$$

For example the second formula follows by taking $\theta = \pi$ and using $(1-\rho)/(1+\rho)\sinh(\gamma)^{-1} = 2\rho/(1+\rho)^2$.

6.1 The dense arithmetic family

Lemma 6.1 (The selected first orders)**.** *Let $p \ge 11$ be prime, choose distinct $a, b \in \{1, \dots, (p-1)/2\}$, and set*

$$\alpha = 2\pi a/p, \quad \beta = 2\pi b/p, \quad E_* = \cos\alpha + \cos\beta = 2\cos\theta, \quad \zeta_* = e^{\mathrm{i}\theta}.$$

For the even family choose $\alpha, \beta \in (0, \pi/2)$ and put $N_0 = 2p$, $k = N_0^2/2 - 1$. For the odd family choose both angles in $(0, \pi/2)$ or both in $(\pi/2, \pi)$ and put $N_0 = p$, $k = (p^2-1)/2$. The first order of the respective parity satisfying a Nickel equation is N_0, and its unordered cosine pair is uniquely $\{\cos\alpha, \cos\beta\}$. No endpoint configuration is resonant at any order. The even family is dense in the upper-right arc; the odd families are dense in both upper quadrants.

Proof. Let $\omega = e^{2\pi\mathrm{i}/p}$, $d = (p-1)/2$, $u_j = \omega^j + \omega^{-j}$, and $F = \mathbb{Q}(\omega + \omega^{-1})$. Every nonzero-index u_j has trace -1. Thus $v = u_a + u_b$ has trace -2 and is irrational: if rational it would be an integer, whereas its trace would force $v = -2/d$. If $p \nmid n$, then $\mathbb{Q}(\omega) \cap \mathbb{Q}(e^{2\pi\mathrm{i}/n}) = \mathbb{Q}$. Indeed the compositum has degree $\varphi(pn) = \varphi(p)\varphi(n)$, so the intersection has degree one. A lower order of the required parity would contradict this fact.

The only rational relation between $1, u_1, \dots, u_d$ is a multiple of $1 + \sum_j u_j = 0$: lift the relation to a polynomial of degree at most $p-1$ and use the minimal polynomial $1 + X + \cdots + X^{p-1}$. At

order p all cosines are $u_j/2$ or 1; at order $2p$ they are $\pm u_j/2$ or ± 1. An equality of two cosine pairs has at most four nonconstant positions. Since $d \geq 5$, a zero coefficient forces the multiple of the relation to vanish, proving uniqueness.

For $E_* > 0$, an endpoint resonance requires a root of unity ξ with $\xi + \xi^{-1} = v - 2$. Its trace is $-2 - 2d < -2d$, impossible because every conjugate of a root-of-unity trace belongs to $[-2, 2]$. For $E_* < 0$ the possible endpoint equation is $\xi + \xi^{-1} = v + 2$. Direct expansion gives $\mathrm{Tr}_F(v^2) = 4d - 6$. If all conjugates of $v + 2$ were in $[-2, 2]$, every conjugate of v would be in $[-4, 0]$, where $v^2 \leq -4v$. Taking traces would imply $4d - 6 \leq 8$, contrary to $d \geq 5$. Approximating any angle in either relevant half interval by two distinct adjacent prime-grid angles proves density. □

6.2 One analytic tail estimate for both isotropic parities

We use the signed coefficients $\mathcal{F}_N^\sigma(E)$, $\sigma \in \{1, -1\}$, on $\mathbb{H}_N^\sigma = \{\sum \theta_i = a_\sigma \pmod{2\pi}\}$, where $a_+ = 0$ and $a_- = \pi$. Thus, with the independent-coordinate Haar measure $\prod_{i<N} \mathrm{d}\theta_i/(2\pi)$,

$$\mathcal{F}_N^\sigma(E) = \frac{1}{N!} \int_{\mathbb{H}_N^\sigma} \frac{1 + \sigma Z}{1 - \sigma Z} \prod_i \frac{1}{\sinh \gamma_i} \prod_{i<l} P_{il} \prod_{i<N} \frac{\mathrm{d}\theta_i}{2\pi}, \qquad P_{il} = \frac{\sin^2((\theta_i - \theta_l)/2)}{\sinh^2((\gamma_i + \gamma_l)/2)}. \tag{6.6}$$

Here $\cosh \gamma_i = E - \cos \theta_i$, $z_i = e^{-\gamma_i}$ is the interior root, and $Z = \prod_i z_i$. Fix $E_* = 2 \cos \vartheta_* \in (0, 2)$, and approach it with

$$\Im E_\varepsilon \asymp \varepsilon > 0, \qquad \Re E_\varepsilon - E_* = O(\varepsilon^2). \tag{6.7}$$

All constants below may depend on this point and on fixed derivative orders, but never on N or ε. The radial exterior approach in the physical coordinate satisfies (6.7).

Theorem 6.2 (Complete isotropic higher tail)**.** *Let $n_* \geq 2$, and put $k = (n_*^2 - 1)/2$ when n_* is odd and $k = n_*^2/2 - 1$ when n_* is even. For either sign σ,*

$$\sum_{\substack{N \geq n_* + 2 \\ N \equiv n_* \ (2)}} |\partial_E^j \mathcal{F}_N^\sigma(E_\varepsilon)| \leq C_j (1 + |\log \varepsilon|), \qquad 0 \leq j \leq k. \tag{6.8}$$

Under any fixed holomorphic change of the exterior parameter, the same bound holds for all parameter derivatives through order k.

6.2.1 A deformation of the complete constrained torus

Set $b = \arccos(E_* - 1)$; then $\vartheta_* < b < 2\vartheta_*$. Choose a smooth periodic p, supported in $(0, \pi)$, with $0 \leq p \leq 1$ and $p = 1$ on $[\delta, \pi - \delta]$; put $m(\theta) = p(-\theta)$. The endpoint width δ is small, so both $\pm b$ lie in the corresponding plateaus. With a small fixed $\tau > 0$, define

$$P = \sum_i p(\theta_i), \quad Q = \sum_i m(\theta_i), \quad h_i = \frac{\tau}{N}(P m(\theta_i) - Q p(\theta_i)), \quad y_i(\xi) = e^{i\theta_i + \xi h_i}, \quad 0 \leq \xi \leq 1. \tag{6.9}$$

The sum of the h_i is zero, so the constrained product of the y_i is unchanged. For $y = e^{h + i\theta}$,

$$\Im \left(E - \frac{y + y^{-1}}{2} \right) = \Im E - \sinh h \sin \theta.$$

The motion is inward on the upper semicircle and outward on the lower one. Its imaginary part therefore remains positive throughout the homotopy. All roots stay on the interior sheet, all

one-body and pair denominators stay nonzero, and $|Z| < 1$. The complete holomorphic integrand can consequently be transported over this closed constrained torus. No partition is imposed before the homotopy.

Write $w_i = \theta_i - ih_i$ at its endpoint, and $M = I - iDh$. Because $\mathbf{1}^T M = \mathbf{1}^T$, the full determinant of M equals its tangent determinant on the constrained hyperplane. Moreover

$$Dh = \frac{\tau}{N}\operatorname{diag}(Pm_i' - Qp_i') + \frac{\tau}{N}(m \otimes p' - p \otimes m').$$

Each summand has operator norm $O(\tau)$, independently of N. Thus M^{-1} is uniformly bounded, $|\det M| \le C^N$, and its fixed angular jets cost at most $C_j^N N^{C_j}$, by the determinant and inverse-matrix differentiation rules.

Let $\eta = 0$ on $(-\infty, 1/4]$ and $\eta = 1$ on $[1/2, \infty)$, with $0 \le \eta \le 1$. On the final contour use

$$\omega_F = \eta(P)\eta(Q), \quad \omega_L = \tfrac{1}{2}(1 - \eta(P))(1 + \eta(Q)), \quad \omega_U = \tfrac{1}{2}(1 - \eta(Q))(1 + \eta(P)).$$

They sum to one. Reflection of all angles exchanges P, Q, sends w_i to $-w_i$, and preserves every complete pair and one-body factor. It also preserves either signed constraint. Consequently

$$\mathcal{F}_N^{\sigma} = F_N + 2L_N. \tag{6.10}$$

The support of L_N has $P \le 1/2$ and contains only the lower semicircle and short upper endpoint slivers. Lift the latter near π across $-\pi$; the rational pair factors and the constraint are periodic. Its angles then lie in $(-\pi - \delta, \delta)$.

6.2.2 The mixed branch populations

On the support of F_N, $P, Q \ge 1/4$. Since the bump supports stay away from $0, \pi$, the imaginary-part identity gives

$$-\log|Z| \ge c\left(N\varepsilon + \frac{\tau PQ}{N}\right) \ge c/N. \tag{6.11}$$

Every true lower branch has normal displacement $\tau P/N$, and every true upper branch has displacement $\tau Q/N$. The square-root form of γ at $W = 1$ then gives a branch distance at least c/N in the W plane and $|\partial_E \log z_i| \le C\sqrt{N}$. These statements persist on a common disk $|E - E_\varepsilon| \le c/N^3$: the variation of $\log Z$ is at most $CN^{3/2}c/N^3$, smaller than half the gap in (6.11). Constants can be decreased once to cover all $N \ge 2$.

Here we use the scalar matching identity. Put $A(u, v) = (u - v)/(1 - uv)$. Subtracting the two dispersion equations gives the exact identity $P_{il} = A(z_i, z_l)A(y_i, y_l)$. The scalar Schur–Pfaffian identity expands the product of the $A(y_i, y_l)$ as a sum over matchings when N is even. When N is odd, append the constant ghost entry obtained by setting its variable to -1; its entries are the removable constant one. In each term combine a matched angular factor with its root factor to obtain the complete P_{il}. Every remaining root factor has modulus at most one on the actual contour, since

$$|1 - uv|^2 - |u - v|^2 = (1 - |u|^2)(1 - |v|^2) + 4\Im u \Im v \ge 0 \tag{6.12}$$

in the closed lower half disk. This is the same scalar matching identity as in Lemma 4.1, with no strict contraction required here.

Let B_0 count coordinates in fixed neighborhoods of the two true branches. Each contributes one to $P + Q$, so $B_0 \le P + Q$. Same-branch complete pairs are bounded: the even inverse in the branch coordinate cancels $\gamma_i + \gamma_l$, leaving $\gamma_i - \gamma_l$. All pairs away from opposite branches are also

bounded by compactness. Opposite-branch coordinates lie in a common sector of width less than π, and hence

$$|1 - z_i z_l| \geq c\sqrt{(P+Q)/N}, \qquad |P_{il}| \leq CN/(P+Q) \leq CN/B_0.$$

At most $B_0/2$ matched pairs are exceptional. Since $B_0 \log(N/B_0) \leq N/e$, their total extra cost is C^N. There are $(N-1)!!$ matchings in the even case and $N!!$ in the odd case. Thus both parities satisfy

$$\left| \prod_{i<l} P_{il} \right| \leq C^N N^{(N+1)/2}. \tag{6.13}$$

This bound holds on the protected complex disk as well. Indeed

$$\partial_E A(u,v) = \frac{u_E(1-v^2) + v_E(u^2-1)}{(1-uv)^2}.$$

Near the branches, $|1-uv| \asymp |\gamma_i| + |\gamma_l|$, $|u_E| \leq C/|\gamma_i|$, and $|\gamma_i|, |\gamma_l| \geq cN^{-1/2}$. The displayed derivative is $O(N)$; outside these neighborhoods it is no larger. Each unmatched root entry therefore grows by at most $O(N^{-2})$ on the c/N^3 disk. Their $O(N^2)$-fold product costs only a constant. The canceled matched pairs retain their previous bounds there by the same sector estimates.

The one-body factors have the independent majorant

$$f(\theta) = C\left(1 + \operatorname{dist}(\theta, \{b, -b\})^{-1/2}\right).$$

Near a branch $\theta_b \in \{b, -b\}$, use the imaginary part of $W - 1$ when $|\theta - \theta_b| \leq C(\varepsilon + |h|)$, and its linear real part outside that region. The small complex disk is absorbed by the branch displacement. In addition, the uniform branch distance c/N gives $|\sinh \gamma_i|^{-1} \leq C\sqrt{N}$ for every coordinate throughout the mixed support and its protected disk; outside the true branch neighborhoods the factor is bounded by a constant. Eliminate the fixed Nth angle with the constraint, bound its one-body factor by $C\sqrt{N}$, and use the independent L^1 majorants for the others. This gives, for every $N \geq 2$,

$$\int_{\mathbb{H}_N^\sigma} \omega_F \prod_i |\sinh \gamma_i|^{-1} \, d\mu_N \leq C\sqrt{N}\, \|f\|_1^{N-1} \leq C^N \sqrt{N}.$$

The estimate remains valid when the harmless cutoff is bounded by one and the remaining independent integrations are enlarged to full circles. The product kernel costs CN by (6.11), and the angular Jacobian costs C^N. Cauchy's inequality on the protected disk and $N! \geq (N/e)^N$ give

$$|\partial_E^j F_N(E_\varepsilon)| \leq C_j^N N^{-N/2+C_j}, \qquad N \geq 2. \tag{6.14}$$

For each fixed $N = 2, 3$, the displaced branches and product gap instead give a bounded analytic integrand on the compact mixed support. Every fixed derivative of these pieces is bounded.

6.2.3 The regular inverse and a cell Gaussian bound

Write $z = e^{-i\varphi}$, $\gamma = i\varphi$, and set $v = \varphi + w$.

Lemma 6.3 (One-sided inverse)**.** *There is a fixed complex neighborhood of the limiting one-sided curve on which $w = U(E, v)$ is analytic with bounded fixed joint jets. On a fixed convex neighborhood $\mathcal{D}_R$ of the right v-interval, including a short negative branch segment, $\Re U_{vv} \geq c > 0$. All the actual one-sided images, including the endpoint slivers, lie in these neighborhoods when the fixed geometric choices are small.*

Proof. On $[-b,0]$, $g(\theta)=\arccos(E_*-\cos\theta)$ has $g'=-\sin\theta/\sin g\geq 0$, and $g'(0)=0$. Thus $v=g(\theta)+\theta$ is strictly increasing, also on a short upper sliver. At the branch, $w=-\arccos(E-\cos\varphi)$ is even analytic in φ, and $\mathrm{d}(w+\varphi)/\,\mathrm{d}\varphi=1$ at zero. The explicit right inverse is

$$U(E,v)=\frac{v}{2}-\arccos\frac{E}{2\cos(v/2)}. \tag{6.15}$$

Its limiting real interval is strictly inside $(-2\vartheta_*,2\vartheta_*)$, because $b<2\vartheta_*$. With $x=v/2$ and $D=4\cos^2 x-E_*^2>0$, differentiation gives

$$U_{vv}(E_*,v)=\frac{E_*}{4\cos^2 x\sqrt{D}}+\frac{E_*\sin^2 x}{D^{3/2}}>0.$$

Compactness preserves a positive real part on a thin convex rectangle. On the left arc $\varphi=-i\gamma$, $\gamma>0$; the real part of v is θ, and $1+\varphi_w$ cannot vanish because φ_w is purely imaginary. The left and right embedded arcs meet in the regular branch chart. Their local analytic inverses consequently patch on a sufficiently small neighborhood of the compact arc. Short endpoint extensions preserve injectivity. Square-root continuity near the branch, and ordinary continuity elsewhere, then put every sufficiently small deformed image inside the chosen domains. □

Put $\phi(E,v)=v-U(E,v)$. With the square-root branch fixed by $\sqrt{4\cos^2(v/2)-E^2}=\sin\phi-\sin U$, implicit differentiation gives

$$\begin{aligned} U_v&=\frac{\sin\phi}{\sqrt{4\cos^2(v/2)-E^2}},\quad U_E=\frac{1}{\sqrt{4\cos^2(v/2)-E^2}},\\ a(E,v)&:=\frac{U_v}{\sinh(i\phi)}=\frac{-i}{\sqrt{4\cos^2(v/2)-E^2}}. \end{aligned} \tag{6.16}$$

The denominator is separated on the fixed inverse neighborhoods. The complete pair is analytic there, including at the double branch: the dispersion identity also writes it locally as

$$P(v_i,v_l)=-\frac{\sin^2((\phi_i-\phi_l)/2)}{\sin^2((U_i+U_l)/2)},$$

whose denominator is nonzero near $U_i=U_l=-b$. Away from that double branch the original pair denominator is separated. The diagonal zero therefore gives the global regular factorization

$$P(v_i,v_l)=(v_i-v_l)^2A(E,v_i,v_l), \tag{6.17}$$

where A and its fixed joint jets are bounded on a smaller neighborhood of the one-sided arc squared.

Lemma 6.4 (Cell contraction and retained collision factors)**.** *On a fixed common complex neighborhood of the one-sided arc, for every fixed ℓ the quotient*

$$\frac{\prod_{i<l\leq N}P_{il}}{\prod_{1\leq i<l\leq\ell}(v_i-v_l)^2}$$

is analytic, and each of its fixed joint jets is bounded by $C_{r,\ell}\exp(-\kappa N^2+C_{r,\ell}N)$ for $N\geq\ell$. The same bound holds after multiplying by $\prod_i a(E,v_i)$. It applies on the enlarged right rectangle with arbitrary actual occupancy fractions in $[0,1]$.

Proof. On the limiting arc $-\pi \le \theta \le 0$, both $y = e^{i\theta}$ and z are in the closed lower half disk: z is on its unit semicircle for $-b \le \theta \le 0$ and on $(0, 1]$ on the remaining segment. Equations (6.12) and $P = A(z_i, z_l)A(y_i, y_l)$ give $|P| \le 1$. Apparent endpoint quotients are filled using the analytic complete pair. Since $P(v, v) = 0$, finitely many cells, say K, cover the compact regular arc so that same-cell pairs have $|P| \le \eta < 1$. Choose a small common complex neighborhood in (E, v): same-cell pairs then have modulus at most $\eta' < 1$, and all pairs have modulus at most $M = 1 + o(1)$. For populations $N_1, \dots, N_K$,

$$\sum_a \binom{N_a}{2} \ge \tfrac{1}{2}(N^2/K - N), \qquad \prod_{i<l} |P_{il}| \le M^{\binom{N}{2}} (\eta'/M)^{(N^2/K - N)/2}.$$

The neighborhood can be fixed small enough that the coefficient of N^2 in its logarithm is negative. This proves $\exp(-\kappa N^2 + CN)$, including on slightly larger fixed polydisks.

Factor the internal ℓ pairs by (6.17). They cost a fixed constant, their cross pairs with the remaining coordinates cost $C^{\ell(N-\ell)}$, and the remaining complete pair product retains its Gaussian bound. Cauchy's inequality on the fixed polydisks gives all asserted joint jets. The bounded one-body product adds only C^N.

The same neighborhood contains the image of the enlarged right rectangle used in coarea. Fix the cells and inverse-domain margins on the undeformed compact arc, then choose the endpoint slivers and negative branch segment inside those margins. In this rectangle every $\rho = P/N$ and $\upsilon = Q/N$ is in $[0, 1]$, and $|h_i| \le \tau$. Away from the branch the change in v_i is $O(\varepsilon + \tau)$; near it the uniform square-root bound makes the change $O(\sqrt{\varepsilon + \tau})$. Choosing τ and then ε_0 small puts all such images in the same fixed neighborhood. □

6.2.4 The exact current and its joint jets

For $\chi_E(\theta) = (v_i)$ define the scalar

$$J = \det D_\theta \chi_E = \det M \prod_i U_{v,i}^{-1}.$$

The physical density equals Jf, where

$$f = \frac{1}{N!(2\pi)^{N-1}} \frac{1 + \sigma Z}{1 - \sigma Z} \prod_i a(E, v_i) \prod_{i<l} P_{il}. \tag{6.18}$$

This is an equality of scalar coefficients on the original constrained angles:

$$J \prod a = (\det M) \prod (\sinh \gamma_i)^{-1}.$$

No intrinsic complex measure on a new constrained v hyperplane is asserted.

For a regular-coordinate field B put

$$V = -M^{-1}(U_E + U_v B). \tag{6.19}$$

If $\sum_i U_{v,i} B_i = -\sum_i U_{E,i}$, then $\sum_i V_i = 0$. If also $\sum_i B_i = 0$, the operator $\partial_E - V \cdot \nabla_\theta$ freezes $\sum_i \varphi_i$. Indeed $\sum \varphi_i = \sum v_i - \sum w_i$, and $\sum w_i = \sum \theta_i$. The exact scalar current identity is

$$\partial_E(Jb) - \operatorname{div}_\theta(VJb) = J \left(E_E b + \sum_i B_i E_i b + \Big(\sum_i E_i B_i \Big) b \right). \tag{6.20}$$

Here b may already be a parameter-dependent transported amplitude. The commuting regular frame is

$$E_i = \sum_l (M^{-1})_{li} U_{v,i} \partial_{\theta_l}, \qquad E_E = \partial_E + \sum_l (M^{-1} U_E)_l \partial_{\theta_l}.$$

It satisfies $E_i v_l = \delta_{il}$ and $E_E v_l = 0$. Every commutator is a linear combination of the angular derivatives, annihilates all v_l, and hence vanishes. Thus this frame applies also to smooth real cutoffs without analytically continuing them. To verify (6.20), differentiating the coordinate Jacobian and using Jacobi's determinant formula gives $J \operatorname{div}_v B = \partial_E J - \operatorname{div}_\theta(JV)$; the product rule gives the displayed identity. Tangency makes its ambient divergence exactly the constrained one, since

$$\sum_{i<N} (\partial_i - \partial_N) V_i = \sum_i \partial_i V_i.$$

Integration by parts is applied to the real and imaginary parts of these complex currents.

Choose one smooth right cutoff r, equal to one for $\theta + b \geq -\delta_B/2$ and zero for $\theta + b \leq -\delta_B$, with its periodic return in the excluded upper plateau. Decompose $1 = \prod_i r_i + (1 - \prod_i r_i)$ on the one-sided term and telescope the second part. Each of its at most N terms has a named compact left anchor a, also on all cutoff jets. It has $|z_a| \leq q_0 < 1$ and $|U_{v,a}| \geq c$. Use

$$B_a = -\frac{\sum_i U_{E,i}}{U_{v,a}}, \qquad B_i = 0 \ (i \neq a). \tag{6.21}$$

This gives tangency; phase freezing is unnecessary because $|1 - \sigma Z| \geq 1 - q_0$. All fixed regular-frame jets of B, V and the cutoffs have polynomial N bounds. The cell Gaussian, the bounded regular one-body factors, and (6.20) then bound every fixed derivative of this whole piece by

$$\frac{C_j N^{C_j}}{N!} \exp(-\kappa N^2 + C_j N). \tag{6.22}$$

In fact, eliminate the compact anchor angle with the constraint. Its Jacobian factor is bounded, and every remaining factor has the independent integrable majorant $C(1 + \operatorname{dist}(\theta, -b)^{-1/2})$. Their integrals cost only C^N. The product kernel and its transported derivatives have polynomial N bounds on this anchored support. This argument applies also at each fixed $N = 2, 3$.

We now work on the full right part, with all angles in $I = [-b - \delta_B, \delta]$. Set $u_i = \theta_i + b$ and, on a diameter shell $r_0 \leq d := \max \theta_i - \min \theta_i \leq 2r_0$, put

$$\mu = N^{-1} \sum_i u_i, \qquad T = |\mu| + r_0 + \varepsilon, \qquad \Delta = r_0/\sqrt{T}. \tag{6.23}$$

These are pointwise envelopes; no derivative of a mean, diameter, or scale is ever taken.

The branch normal form, for a common outward displacement $h \geq 0$, is

$$\begin{gathered} 1 - W = \sin b \, u - i\beta + R, \qquad c(\varepsilon + h) \leq \beta \leq C(\varepsilon + h), \\ R \in \mathbb{R}, \qquad |R| \leq C(u^2 + \varepsilon^2 + h^2), \qquad \varphi = \sqrt{2(1 - W)}(1 + O(1 - W)). \end{gathered} \tag{6.24}$$

Indeed $W = E - \cosh(h + i(-b + u))$ and $\beta = \Im E + \sinh h \sin(b - u)$; its real part has the displayed Taylor expansion. The physical square root lies in a small enlargement of the fourth quadrant. In particular

$$|1 + \varphi_w(w_i)| \leq C(|u_i| + \varepsilon)^{-1/2}, \quad -\Im \varphi_i \geq c\sqrt{|u_i| + \varepsilon} \quad (u_i \leq 0),$$
$$\int_{\mu - 2r_0}^{\mu + 2r_0} (|u| + \varepsilon)^{-1/2} \, \mathrm{d}u \leq C r_0/\sqrt{T}.$$

The interval bound follows by comparability if $|\mu| \ge 4r_0$; otherwise enlarge to $[-6r_0, 6r_0]$ and rationalize $\sqrt{6r_0+\varepsilon} - \sqrt{\varepsilon}$. Joining two angles along their one-variable curve with the actual $\rho = P/N, \upsilon = Q/N$ held fixed gives

$$|v_i - v_l| \le C\Delta, \qquad |U_{v,i}| \le C\sqrt{T}. \tag{6.25}$$

The joining derivative has the additional bounded factor $1 - i\tau(\rho m' - \upsilon p')$. Near the lower branch its outward shift is the common $\tau\rho$, so the preceding interval bound applies; elsewhere it is regular. For the second bound, if $P = 0$ then every actual $h_i = 0$, and the branch form and (6.16) apply directly. If $P > 0$, some actual angle is positive, so $\max u_i \ge b$ and $T \ge b/2$; boundedness of U_v proves the same result. These arguments allow all actual occupancy fractions in $[0, 1]$.

Use the filled analytic divided differences

$$\mathcal{U}_{il} = \int_0^1 U_v(v_l + t(v_i - v_l))\, dt, \qquad \mathcal{E}_{il} = \int_0^1 U_{vv}(v_l + t(v_i - v_l))\, dt.$$

The convex inverse domain gives $\Re\mathcal{E}_{il} \ge c$. Define the analytic field

$$\begin{gathered} D_U = \sum_{i<l} (U_i - U_l)^2, \qquad H_{il} = \frac{(v_i - v_l)\mathcal{U}_{il}^2}{\mathcal{E}_{il}}, \qquad G_i = \sum_l H_{il}, \\ A = -\sum_i U_{E,i}, \qquad B_i = AG_i/D_U. \end{gathered} \tag{6.26}$$

There is no pair pole: H_{il} is analytic also at $v_i = v_l$. Away from equality it equals $(U_i - U_l)^2/(U_{v,i} - U_{v,l})$. Antisymmetry and pairing give

$$\sum_i G_i = 0, \qquad \sum_i U_{v,i} G_i = D_U, \qquad \sum_i B_i = 0, \qquad \sum_i U_{v,i} B_i = -\sum_i U_{E,i}.$$

Thus the field has both exact transport identities.

Lemma 6.5 (Protected analytic variance and joint jets)**.** *On each actual right shell,*

$$\Re D_U \ge c \sum_{i<l} (\theta_i - \theta_l)^2 \ge cT\Delta^2 > 0. \tag{6.27}$$

For every fixed total derivative order s,

$$|\mathcal{D}^s B_i| + |\mathcal{D}^s V_i| \le C_s N^{C_s} \Delta^{-1-s}. \tag{6.28}$$

The field is regular at every partial diagonal.

Proof. On the actual contour $U_i = w_i = \theta_i - ih_i$, and the real balanced shifts satisfy $|h_i - h_l| \le C\tau|\theta_i - \theta_l|$ at fixed actual occupancies. Hence $\Re(U_i - U_l)^2 = (\theta_i - \theta_l)^2 - (h_i - h_l)^2 \ge (1 - C^2\tau^2)(\theta_i - \theta_l)^2$. Small fixed τ proves the denominator bound.

Freeze the actual values of T, Δ and consider the polydisk

$$|\delta E|,\ |\delta v_i| \le c_0 \Delta/N^2.$$

Choose c_0 small so it lies in the fixed inverse neighborhoods. Throughout this disk, $|v_i - v_l| \le C\Delta$, $|U_v| \le C\sqrt{T}$, and $|U_E(v_i) - U_E(v_l)| \le C\Delta$. Indeed the last bound is an analytic divided difference; the middle one follows from its actual endpoint bound and bounded joint jets, since $\Delta \le \sqrt{T}$.

Along the straight parameter and coordinate segments, the common parameter change therefore moves each $U_i - U_l$ by $O(\Delta|\delta E|)$, and the two coordinate changes move it by $O(\sqrt{T}(|\delta v_i| + |\delta v_l|))$. Their total change is at most $Cc_0 r_0/N^2$. Since each original difference is $O(r_0)$, summing the $N(N-1)/2$ squared differences gives

$$|D_U^{\mathrm{new}} - D_U| \le Cc_0 r_0^2 \le \tfrac{1}{2}\Re D_U.$$

Thus $|D_U^{\mathrm{new}}| \ge cr_0^2$ on this whole polydisk. Also $|\mathcal{U}_{il}| \le C\sqrt{T}$ and $|\mathcal{E}_{il}| \ge c$ there, so $|H_{il}| \le Cr_0\sqrt{T} = CT\Delta$, $|A| \le CN$, and $|B_i| \le CN^2/\Delta$. Cauchy's inequality now gives all fixed joint jets at once:

$$|\partial_E^a \partial_v^\beta B_i| \le C_s N^{2+2s} \Delta^{-1-s}, \qquad s = a + |\beta|. \tag{6.29}$$

The commuting-frame coefficients $M^{-1}\operatorname{diag} U_v$ and $M^{-1}U_E$ have polynomially bounded fixed joint jets: the real angular jets of M and analytic jets of U are bounded, and differentiating M^{-1} inductively bounds the frame jets. The frame formulas use U_v with a nonnegative power. The same induction bounds the fixed smooth cutoff jets. Formula (6.19), the product rule, and (6.29) give the same field bound for V. The denominator is separated off full equality, including at partial diagonals where the analytic quotients H_{il} extend. □

Fix now $\ell = n_* + 1$ and $D_\ell = \ell(\ell-1)$. Factor the first ℓ collision variables using Lemma 6.4. For the initial smooth weighted regular amplitude b_0 after removing the product kernel,

$$|\mathcal{D}^s b_0| \le \frac{C_s N^{C_s}}{N!} e^{-\kappa N^2 + C_s N} \Delta^{D_\ell - s}, \qquad s \le 2k.$$

The same statement includes parameter derivatives, which preserve the explicit collision polynomial. Iterate the exact current with

$$b_{j+1} = \mathcal{T} b_j, \qquad \mathcal{T} b = E_E b + \sum_i B_i E_i b + \Big(\sum_i E_i B_i\Big) b. \tag{6.30}$$

This recurrence differentiates every parameter-dependent descendant coefficient and cutoff. Because the field freezes $\sum \varphi_i$, the product kernel is unchanged. Induction using (6.28) gives

$$|\mathcal{D}^s b_j| \le \frac{C_j N^{C_j}}{N!} e^{-\kappa N^2 + C_j N} \Delta^{D_\ell - 2j - s}, \qquad s \le 2(k-j). \tag{6.31}$$

Indeed E_E loses at most one scale power, while $B_i E_i$ and $(E_i B_i)$ each lose at most two. Leibniz' rule gives the same count for joint jets. Only polynomially many index choices occur at each fixed order.

At fixed $N, \varepsilon > 0$ puncture the finitely many full-equality points by angular balls of radius r. r. All geometric Jacobians and the product kernel are then smooth; $V = O(r^{-1})$, and the retained collision degree is D_ℓ. The flux after j transports is bounded by a constant times $r^{D_\ell + N - 3 - 2j}$, which tends to zero for every $N \ge n_* + 2$ and $j \le k$. Partial diagonals require no puncture by (6.27). The original smooth weights vanish at the outer chart boundaries, and diameter cutoffs are introduced after differentiation. Thus the jth derivative of the complete right term is the integral of its transported density, bounded by

$$\frac{C_j N^{C_j}}{N!} e^{-\kappa N^2 + C_j N} \Delta^{D_\ell - 2j} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - \sigma Z|}. \tag{6.32}$$

6.2.5 Coarea with coupled occupancies

The phase derivative for (6.32) includes the derivatives of both occupancies.

First, for $G = x - ia$, $a \ge a_0 > 0$, and fixed $|\vartheta| < \pi/2$, put $\nu = \Re(e^{-i\vartheta}G)$ and $P_\vartheta = 2\pi\cos\vartheta$. If $a_0 \le 1$, then

$$\frac{1}{|1 - e^{-iG}|} \le \frac{C_\vartheta}{a_0 + \operatorname{dist}(\nu, P_\vartheta\mathbb{Z})}. \tag{6.33}$$

For $a \le 1$, this follows from $|1 - e^{-a-ix}| \ge c(a + \operatorname{dist}(x, 2\pi\mathbb{Z}))$ and $|\nu - 2\pi n\cos\vartheta| \le \cos\vartheta|x - 2\pi n| + |\sin\vartheta|a$. For $a \ge 1$, the left denominator is separated and the proposed right denominator is bounded above. Integrating the majorant over an interval of length L costs at most $C_\vartheta(1 + L)(1 + |\log a_0|)$. For $\sigma = -1$ replace G by $G - \pi$. On the actual contours $-\log|Z| \ge cN\varepsilon$: write $\varphi_i = x_i - ia_i$, and use $\Im W_i = \sin x_i \sinh a_i \le Ca_i$ on the bounded phase range. Apply the lemma with $a_0 = \min(1, cN\varepsilon)$. Its logarithmic cost is at most $C(1 + L)(1 + |\log\varepsilon|)$.

Lemma 6.6 (Complete right-shell coarea)**.** *On each fixed-mean lift of the full right rectangle $I^N \cap \mathbb{H}_N^\sigma$, with the actual balanced occupancies, the integral over any diameter shell in* (6.23) *obeys*

$$\int_{\substack{\sum u_i = N\mu \\ r_0 \le d \le 2r_0}} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - \sigma Z|} \prod_{i<N} \mathrm{d}\theta_i \le C^N N^C \Delta^{N-3}(1 + |\log\varepsilon|). \tag{6.34}$$

There are at most CN constrained means. The constants are uniform in each of them; their subsequent sum costs only this polynomial factor.

Proof. Write $m_0 = \min u_i$, $M_0 = \max u_i$, and first suppose $0 \le m_0 < M_0$. With other angles fixed, the constraint fixes $m_0 + M_0$. Put $G = \sum_i \varphi_i$ and $\vartheta = 3\pi/8$. We claim

$$\Re(e^{-i\vartheta}\,\mathrm{d}G/\,\mathrm{d}m_0) > 0, \qquad \frac{|(1 + \varphi_w(w_{m_0}))(1 + \varphi_w(w_{M_0}))|}{\Re(e^{-i\vartheta}\,\mathrm{d}G/\,\mathrm{d}m_0)} \le C\frac{\sqrt{M_0 + \varepsilon}}{M_0 - m_0}. \tag{6.35}$$

For the one-variable comparison curve, fix the values $\rho = P/N, \upsilon = Q/N$ at the configuration and set $f(u) = -b + u - i\tau\{\rho m(-b + u) - \upsilon p(-b + u)\}$, $E = \varepsilon + \tau\rho$. For $u \ge 0$, the branch normal form and compactness on the rest of I give

$$|\varphi_w(f(u))| \le C(u + E)^{-1/2}, \qquad |\varphi_{ww}(f(u))| \asymp (u + E)^{-3/2}.$$

The inverse supplies $-\varphi_{ww} = U_{vv}/U_v^3$. For $u \ge 0$ the argument of U_v is in a small enlargement of $[-\pi/4, 0]$, while U_{vv} has small argument. Hence $-\varphi_{ww}$ has uniformly positive $3\pi/8$ projection. If $a = f' = 1 + O(\tau)$ and $B(u) = \varphi_w(f(u))a(u)$, then $-B' = -\varphi_{ww}(f)a^2 - \varphi_w(f)a'$ retains that projection for small τ. Consequently

$$\begin{gathered} \Re\{e^{-i\vartheta}[B(m_0) - B(M_0)]\} \ge cL, \\ L = \int_{m_0}^{M_0} (u + E)^{-3/2}\,\mathrm{d}u = \frac{2(M_0 - m_0)}{\sqrt{m_0 + E}\sqrt{M_0 + E}(\sqrt{m_0 + E} + \sqrt{M_0 + E})}. \end{gathered} \tag{6.36}$$

Now restore all occupancy derivatives. Put $B_m^{\mathrm{occ}} = \sum_l m(\theta_l)\varphi_w(w_l)$ and $B_p^{\mathrm{occ}} = \sum_l p(\theta_l)\varphi_w(w_l)$. Direct differentiation of (6.9) gives the exact identity

$$\partial_{\theta_i} G = \varphi_w(w_i)[1 - i\tau(\rho m_i' - \upsilon p_i')] - \frac{i\tau p_i'}{N} B_m^{\mathrm{occ}} + \frac{i\tau m_i'}{N} B_p^{\mathrm{occ}}. \tag{6.37}$$

Every actual $u_l \ge m_0$, so $(|B_m^{\mathrm{occ}}| + |B_p^{\mathrm{occ}}|)/N \le C(m_0 + E)^{-1/2}$. Taking $\partial_{\theta_{\min}} - \partial_{\theta_{\max}}$, both bump-derivative differences are $O(M_0 - m_0)$. The entire occupancy correction is therefore at most

$C\tau(M_0-m_0)(m_0+E)^{-1/2}\le C\tau L$, by the exact expression for L and the fixed upper bound on M_0+E. Small fixed τ absorbs it into the positive projection. The numerator of (6.35) is at most $C[(m_0+E)(M_0+E)]^{-1/2}$, giving ratio $C\sqrt{M_0+E}/(M_0-m_0)$. If $P=0$, $E=\varepsilon$; if $P>0$, an actual positive angle gives $M_0\ge b$, and hence $E\le C(M_0+\varepsilon)$. This proves (6.35) with all actual occupancies, including on the enlarged rectangle.

After transport, bound every smooth coefficient by its proved envelope and enlarge to this full rectangle. All its occupancy fractions lie in $[0,1]$, the balanced displacement preserves $\Im W_i\ge\Im E$, and Lemma 6.4 has already placed its images in the regular domains. Fix a constrained mean $\mu=b+(2\pi l+a_\sigma)/N$ and the two extreme indices; at most CN^3 such choices occur. Eliminate the maximum coordinate with the constraint. With the remaining $N-2$ coordinates fixed, conditions specifying the extrema, the shell, and a nonnegative minimum are linear interval conditions in m_0, with $M_0=A-m_0$. Thus the fiber is one interval, and its projected phase is injective by (6.35). Its phase range is at most CN. Equations (6.33) and (6.35) bound this extreme-coordinate integral by

$$CN\frac{\sqrt{T}}{r_0}(1+|\log\varepsilon|).$$

If $m_0<0$, its branch attenuation from (6.24) instead gives the direct bound

$$\frac{|(1+\varphi_w(w_{m_0}))(1+\varphi_w(w_{M_0}))|}{|1-\sigma Z|}\le\frac{C}{(|m_0|+\varepsilon)\sqrt{|M_0|+\varepsilon}}.$$

With $a=|m_0|+\varepsilon$ and $c=|M_0|+\varepsilon$, $a+c\ge r_0$, and

$$\frac{1}{a\sqrt{c}}\le\frac{1}{\sqrt{r_0}}\left(\frac{1}{a}+\frac{1}{c}\right).$$

Each reciprocal integrates to $C(1+|\log\varepsilon|)$ on the bounded slice $M_0=A-m_0$. Since $T\ge r_0$, this gives the same extreme pair envelope $C\sqrt{T}/r_0(1+|\log\varepsilon|)$. Only now replace the other $N-2$ Jacobian factors by the independent majorants and integrate each over $[\mu-2r_0,\mu+2r_0]$. Their total cost is $(Cr_0/\sqrt{T})^{N-2}$. Combining the extreme pair, the ordinary factors, and the polynomial number of cells proves (6.34). □

6.2.6 One sum for all higher particle orders

Combining (6.32) with Lemma 6.6 bounds a right shell at a fixed mean by

$$\frac{C_jN^{C_j}}{N!}e^{-\kappa N^2+C_jN}\Delta^{D_\ell+N-3-2j}(1+|\log\varepsilon|). \tag{6.38}$$

Its exponent is positive for every $N\ge n_*+2$ and $j\le k$: it is at least n_*+N-2 in the odd case and n_*+N-1 in the even case. For fixed μ,ε,

$$\frac{\Delta(r_0/2)}{\Delta(r_0)}=\frac{1}{2}\sqrt{\frac{|\mu|+r_0+\varepsilon}{|\mu|+r_0/2+\varepsilon}}\in[1/2,1/\sqrt{2}].$$

The dyadic sum is therefore bounded uniformly in the mean by a geometric series. Its largest possible scale is fixed; its power costs only C^N times a fixed constant. This is absorbed into the linear exponential in (6.38). Summing the at most CN means adds only a polynomial factor. No

small/large-scale division or finite intermediate particle range is required. The mixed, anchored, and right pieces consequently obey, for all higher orders,

$$|\partial_E^j \mathcal{F}_N^\sigma(E_\varepsilon)| \le C_j(1+|\log\varepsilon|)\left(C_j^N N^{-N/2+C_j} + \frac{N^{C_j}}{N!}e^{-\kappa N^2+C_j N}\right). \tag{6.39}$$

Both envelopes are summable. The exact split (6.10) proves Theorem 6.2. The parameter-change assertion follows from the finite chain rule.

Proposition 6.7 (Transport estimates at fixed particle order)**.** *For each fixed $N \ge 2$ and every fixed derivative order, the mixed piece is bounded and extends holomorphically across E_* on its deformed contour. A smooth one-sided term with a compact left anchor has bounded derivatives of every fixed order. A smooth right-supported term whose angular diameter is bounded below by a fixed $d_0 > 0$ has derivatives $O(1+|\log\varepsilon|)$ of every fixed order. The assertions allow arbitrary fixed smooth angular cutoffs with those supports.*

Proof. For the fixed mixed order, both true branches and the product denominator have uniform gaps on the compact support, as proved above; the same small parameter disk works for the whole integrand. This includes $N = 2, 3$ directly. The anchored field (6.21) and its integrable independent Jacobian majorants prove the second assertion at any fixed derivative budget. On a right support with $d \ge d_0$, all jets of the covariance field are bounded and there is no collision puncture. Iterate (6.30) with that budget, and then apply Lemma 6.6; the finitely many dyadic scales bounded away from zero cost $O(1+|\log\varepsilon|)$. For $N = 2$ the coarea keeps both extreme factors and has zero remaining one-body integrations, so the same argument applies. All differentiated cutoff factors are included in the exact recurrence. □

For negative real dispersion in the odd family, the exact identity (6.3), followed by conjugation, sends the exterior approach to the signed positive geometry. More explicitly, for $\widehat{\zeta} = -\overline{\zeta}$ and odd N,

$$(C_N^+)^{(j)}(\zeta) = (-1)^{j+1}\overline{(C_N^-)^{(j)}(\widehat{\zeta})}.$$

Thus Theorem 6.2 controls every higher physical odd coefficient also at the original negative-dispersion points. The sign changes both the momentum constraint and the product kernel.

6.3 The finite complement and the original prime family

Lemma 6.8 (All fixed derivatives of the finite complement)**.** *At every point of Lemma 6.1, all fixed derivatives of each smaller coefficient of the selected parity are bounded along $\zeta_\varepsilon = (1+\varepsilon)\zeta_*$. At the first order N_0, choose small fixed smooth shape cutoffs at the four angular configurations $q_i = \alpha, -\alpha, \beta, -\beta$ for every i. The complete coefficient minus these four local integrals has bounded derivatives of every fixed order on the same approach.*

Proof. First suppose $E_* > 0$. For the finitely many relevant orders choose the branch width smaller than one quarter of the distance from the branch to any constrained equality mean. This distance is positive by the endpoint exclusion in Lemma 6.1. Use the whole-torus deformation and exact reflection split proved above. The mixed term has a fixed product gap and displaced branches at every fixed order $N \ge 2$. Its derivatives are bounded. The compact-left-anchor terms have the same property by Proposition 6.7. The one-particle term is $1/(E-2)$, and is regular here.

At equality on the right support, the balanced displacement vanishes: $P = Np(q_0)$, $Q = Nm(q_0)$ give $Pm(q_0) - Qp(q_0) = 0$. All roots and complete pairs are regular at these finitely many points. A vanishing product denominator requires

$$E_* = \cos q_0 + \cos \varphi_0, \qquad Nq_0, N\varphi_0 \in 2\pi\mathbb{Z}.$$

The arithmetic excludes such a point at every smaller order and leaves exactly $q_0 = -\alpha, -\beta$ in the first lower chart. Their reflected configurations give the two upper charts. Every other equality neighborhood has a fixed product gap. Choose the critical cutoffs inside the lower plateau, where all $m_i = 1, p_i = 0$: the deformation is zero there and the lower weight and the right cutoff are one.

After these neighborhoods are removed, the remaining right support has angular diameter at least a fixed $d_0 > 0$. The finite-order The finite-order transport estimate gives $O(1 + |\log \varepsilon|)$ for each fixed derivative on this support, including all cutoff derivatives and all remaining phase zeros. This gives the same bound at every fixed order for each complete lower coefficient and for the complete remainder at the first singular order. These remainders are holomorphic on the one-sided parameter sector: their contours and smooth weights are independent of E. For each fixed j, integration of the next derivative gives

$$\partial_E^j R(E(\zeta_\varepsilon)) = \partial_E^j R(E(\zeta_{\varepsilon_1})) - \int_\varepsilon^{\varepsilon_1} \frac{dE(\zeta_u)}{du} \partial_E^{j+1} R(E(\zeta_u))\, du.$$

The logarithmic bound is integrable and $dE(\zeta_u)/du$ is bounded. Every stated derivative is therefore bounded. The finite chain rule gives the conclusion in ζ.

For $E_* < 0$ in the odd family, apply conjugation and (6.3) to obtain the signed positive geometry at $\widehat{E}_* = -E_* > 0$. The transformed constraint and product kernel both have sign -1. Equality resonance is carried exactly to the original resonance by this invertible transformation; the endpoint exclusion is also unchanged. The preceding argument uses either sign, so it applies to this complete transformed remainder. The inverse transformation gives the bound on the original odd sum. □

Proposition 6.9 (The complete first amplitude)**.** *Let $N = N_0$, $a_N = (N^2 - 1)/2$, $k = \lfloor a_N \rfloor$ and $\nu = k + 1 - a_N \in \{1/2, 1\}$. With the original angles and exterior parameter of Lemma 6.1,*

$$C_N^{(k)}(\zeta_\varepsilon) = 4L_* \varepsilon^{-\nu} + o(\varepsilon^{-\nu}), \qquad L_* \neq 0.$$

All derivatives below k are bounded for even N. For odd N they are bounded through $k - 2$ and at most logarithmic at $k - 1$. In particular these are the same four full-coefficient chart contributions for the distinct-angle family in both signs of E_*.

Proof. In the lower chart put $q_i = -\alpha + u_i$, $\sum_i u_i = 0$, $Q_2(u) = \sum_i u_i^2$ and $\Delta_N(u) = \prod_{i<j}(u_i - u_j)$. The physical branch has $\gamma_i = i\varphi_i$ and $\varphi_i = \beta$ at the center. Implicit differentiation of $\cos\varphi = E - \cos q$ gives

$$\varphi_q = \frac{\sin\alpha}{\sin\beta}, \qquad \varphi_{qq} = -\frac{H_{\alpha\beta}}{\sin^3\beta} = -2d, \quad H_{\alpha\beta} = E_*(1 - \cos\alpha\cos\beta), \quad d = \frac{H_{\alpha\beta}}{2\sin^3\beta} \neq 0.$$

Put $Q = 2N\sin\theta/\sin\beta > 0$. The literal angular constraint cancels the linear shape term and its mixed parameter derivative. Since $N\beta \in 2\pi\mathbb{Z}$, it follows that

$$1 - Z = Q\varepsilon - idQ_2(u) + O\left(\varepsilon^2 + \varepsilon Q_2(u) + Q_2(u)^2 + \sum_i |u_i|^3\right), \tag{6.40}$$

$$\partial_\zeta(1 - Z)(\zeta_*, 0) = Qe^{-i\theta}. \tag{6.41}$$

The leading terms have perpendicular phases. On a fixed sufficiently small real shape ball and for small $\varepsilon > 0$ this gives $|1 - Z| \geq c(\varepsilon + Q_2(u))$.

The local integral is exactly

$$I_N(\zeta) = \int_{\mathbb{R}^{N-1}} \frac{\chi(u)\Delta_N(u)^2 A_N(\zeta, u)}{1 - Z}\, du_1 \cdots du_{N-1}, \qquad u_N = -\sum_{i<N} u_i,$$

with analytic amplitude and

$$A_N(\zeta_*, 0) = \frac{2i^{-N^2}}{N!(2\pi)^{N-1}2^{N(N-1)}\sin^{N^2}\beta}.$$

Here every pair numerator contributes $(u_i - u_j)^2/4$, its denominator contributes $(i\sin\beta)^2$, the N one-body factors contribute $(i\sin\beta)^{-N}$, and $1 + Z$ contributes two. Thus all Haar and factorial normalizations are included.

Define the positive angular constant

$$c_N = \frac{1}{\sqrt{N}} \int_{\substack{\sum v_i = 0 \\ \sum v_i^2 = 1}} \Delta_N(v)^2\, d\sigma(v) > 0.$$

The factor $1/\sqrt{N}$ is the Gram determinant correction for the independent coordinates $v_1, \ldots, v_{N-1}$ on this hyperplane. Polar coordinates, followed by $x = r^2$, give

$$\begin{aligned}\int_{\mathbb{R}^{N-1}} \frac{\Delta_N(v)^2}{(Q - idQ_2(v))^{k+1}}\, dv &= \frac{c_N}{2}\int_0^\infty \frac{x^{a_N - 1}}{(Q - idx)^{k+1}}\, dx \\ &= \frac{c_N}{2} Q^{a_N - k - 1}(-id)^{-a_N}\, \mathrm{B}(a_N, k + 1 - a_N).\end{aligned}$$

For a positive real coefficient in place of $-id$, substitution gives the beta integral. Continue that coefficient through the appropriate quadrant to $-id$. On each such closed quadrant of fixed positive modulus, $|Q + cx| \geq c_0(1 + x)$, so dominated convergence proves the identity also at the endpoint. The radial tails are r^{-2} for even N and r^{-3} for odd N.

In the kth derivative of the local integral the unique highest pole has numerator $(-1)^k k!(\partial_\zeta(1 - Z))^k \chi \Delta_N^2 A_N$. Under $u = \sqrt{\varepsilon} v$ its scale is $\varepsilon^{-\nu}$, and its scaled integrand has the integrable majorant $C\Delta_N(v)^2/(1 + Q_2(v))^{k+1}$. Dominated convergence therefore gives its leading coefficient by the preceding period. Every other differentiated term has a lower pole order and is bounded in the even case and at most logarithmic in the odd case. The same radial estimates prove the lower-derivative statements. The one-chart coefficient simplifies to

$$L_* = \frac{i^{-N^2}(-1)^k k!\, c_N\, \mathrm{B}(a_N, k + 1 - a_N) 2^{a_N}(2N\sin\theta)^{a_N - 1} e^{-ik\theta}}{N!(2\pi)^{N-1}2^{N(N-1)}(-iH_{\alpha\beta})^{a_N}}. \tag{6.42}$$

All sine powers cancel, since $-N^2 - (a_N - 1) + 3a_N = 0$. For even N, $H_{\alpha\beta} > 0$ and the last power uses argument $-\pi/2$. For odd N, $a_N = k$ is divisible by four, so the same power is $|H_{\alpha\beta}|^k$ for either sign. Every factor is nonzero and the expression is symmetric in α, β. Their two lower charts therefore have exactly the same complex coefficient. Reflection of every angle preserves the complete constrained density and gives the two equal upper charts. Lemma 6.8 disposes of their complete complement, proving the factor four and the claimed full-coefficient asymptotic. □

6.4 Dense local obstructions and the exact physical identities

The full-site form-factor identification gives, on the respective real physical intervals,

$$\mathcal{X}_L(s) = (1-s^{-4})^{1/4} \sum_{N\geq 2,\ N \text{ even}} C_N(s), \qquad s>1, \tag{6.43}$$

$$\mathcal{X}_H(s) = \frac{(1-s^4)^{1/4}}{s} \sum_{N\geq 1,\ N \text{ odd}} C_N(1/s), \qquad 0<s<1. \tag{6.44}$$

The low-temperature formula has the plus-phase subtraction already performed: its omitted zero-particle term is M_+^2, and its onsite contribution is $1-M_+^2$. The origin is already included in each full-site coefficient; no additional constant is appended to either complete sum. The high-temperature onsite sum is one, and $C_1 = 1/(E-2)$ checks the normalization. The branches in (6.43)–(6.44) are positive on their stated real intervals.

Proposition 6.10 (Dense obstructions for the physical isotropic sums)**.** *The full normalized even exterior sum in* (6.43) *and the full normalized odd interior sum in* (6.44) *have no local holomorphic extension at a dense set of points of* $|s|=1$*. At their selected points the nonzero leading derivatives are those of Proposition 6.9, multiplied by the physical nonvanishing factors. These assertions use the full-tail bounds of Theorem 6.2.*

Proof. At every selected point, Proposition 6.9, the bounded lower orders, and Theorem 6.2 give a nonzero leading derivative of the complete physical sum. All lower derivatives are smaller than that leading scale, so multiplication by its local analytic nonzero phase prefactor preserves it by the product rule. A holomorphic extension through such a point would bound that fixed finite derivative on the short radial segment, a contradiction. The selected points are dense on the required arcs. For the even sum, the contour transformation $(s,x,y)\mapsto(-s,-x,-y)$ gives evenness, and conjugation completes all four quadrants. For the odd sum both signs were treated explicitly; conjugation completes a dense subset of the circle. The inversion $s=1/\zeta$ is locally biholomorphic at each unit-circle point, so the odd conclusion concerns the physical interior germ. This proves the stated local obstructions. Their passage to all boundary points and their role in the exact slice and isotropic corollaries are established in Section 7. □

The high-temperature value at $s=0$ is the removable value one. Indeed, uniformly in the real angular variables, the decaying transfer root satisfies $\rho = s/2 + O(s^2)$; each one-body factor is $O(s)$ and each squared pair is $O(s^2)$, giving $C_N(1/s) = O(s^{N^2})$ with a bound $C^{N^2}|s|^{N^2}/N!$ for sufficiently small $|s|$. The $N=1$ coefficient is $s/(1-s)^2$, and every higher odd order vanishes after division by s. The exterior normalized low-temperature germ is single valued; the energy-field susceptibility is interpreted on the physical continuation of $\beta_{\text{phys}} = \operatorname{arsinh}(s)/(2J)$. At every selected nonaxis point this factor is analytic and nonzero, so the same dense obstructions hold for $\chi = \beta_{\text{phys}}\mathcal{X}$. On the exterior the inverse temperature is read on the physical cover.

The exact coefficient identities and Lemma 2.13 fix the two full sums on the equal-coupling cells. The reduced LT coefficients depend only on $s=\sinh z$ and are invariant under $z\mapsto z+2\pi ik$; the physical field factor is $\beta = z/(2J)$ on that covering.

7 Proofs of the joint, slice and isotropic theorems

7.1 The complete joint obstruction

Proof of Theorem 1.1. Lemma 2.1 identifies the endpoint locus and contains the entire boundary of the unsquared spectral set in it. Theorem 2.2 identifies the physical components and their

contractibility and p-bounds. Proposition 2.4 gives their plurisubharmonic exhaustion. The physical form-factor and coefficient identities with Proposition 3.2 give joint holomorphy and normal convergence of the physical sums.

Take a regular face point whose free coefficient is $b = \cos(2\pi a/\ell)$, with ℓ an odd prime and $0 < a < \ell/2$. By Proposition 2.7 the variables (C, B), after exchanging transfer axes if needed, are independent holomorphic local coordinates, $\Im B \neq 0$, and the physical domain is the full complementary side of that edge. At such a point $S, T \neq 0$, and the edge identity (2.15) implies $p^2 \neq 1$. Thus the locally chosen physical prefactor is analytic and nonzero.

At a positive endpoint, restrict to any holomorphic transversal with nonramified edge coordinate $u = C - B - b$, and write its anisotropy as $B(u)$. Conjugating the data when needed gives $\Im B(0) > 0$, and the physical side contains $u = \mathrm{i}\varepsilon$. Proposition 4.3 proves a nonzero singular derivative of the entire odd or even sum, including the complete differentiated higher tail and physical prefactor. The same proposition handles a negative endpoint for the even sum by the exact angular shift and, if necessary, conjugation.

For a negative endpoint of the odd sum use

$$t = \frac{C + B - b}{B}.$$

This is also nonramified on the stipulated transversal, because $B(0) \neq 0$. In the lower-half-root orientation the anisotropy $B(t)$ has $\Im B(0) < 0$, and the local physical side is $\Im(B(t)t) > 0$. Theorem 5.1 gives all finite boundary jets of the complete odd sum and a selected sequence whose coefficient roots diverge. Its exact full-sum asymptotic (5.66) excludes cancellation by every other particle index and winding. Division by the nonvanishing physical prefactor proves the same nonextension for the physical response. The opposite orientation follows by conjugation. Sections 4 and 5 give the estimates uniformly on the indicated compact complex-parameter families.

In particular choose a constant-B transversal. If the joint physical function extended to a neighborhood of the selected point, its restriction to that disk would give the forbidden one-variable extension. The selected points are dense on regular faces by Proposition 2.7, and regular faces are dense in the entire physical boundaries by Proposition 2.6. Corollary 2.12 then excludes componentwise extension at every boundary point, including corners, collinear points and vanishing transfer-coordinate charts. On the collinear equal-coupling restriction the local obstructions are those of Proposition 6.10.

Finally consider analytic continuation in an unramified Riemann domain over $\mathbb{C}^2$ from one of these physical germs. As long as the projected continuation path remains in its domain, uniqueness of analytic continuation makes every continued germ the germ of the single-valued function already constructed there. If a path first leaves that domain, its first exit is a boundary point. A local biholomorphic chart of the Riemann domain at the continued point would yield a holomorphic function on a base neighborhood agreeing with the physical function on an accumulating domain-side component. The proved componentwise obstruction forbids this. Compactness of the finite path interval guarantees a finite first exit whenever the path reaches outside. The obstruction therefore holds on every unramified continuation reached from the physical germ. □

7.2 All physical ratios, including the collinear one

Lemma 7.1 (Complex neighborhoods of the coupling ratio)**.** *At a regular selected point on a positive-ratio slice with nonzero edge derivative, the transverse representation is holomorphic in the complex ratio λ nearby. The boundary estimates hold uniformly on common compact parameter polydisks.*

Proof. The boundary temperature coordinate z_0 is nonzero. Thus $(z, \lambda) \mapsto (z, w = \lambda z)$ has determinant z_0 and is a local biholomorphism. Its composition with the regular (C, B) coordinate map is holomorphic. If the positive-edge parameter is $u = C - B - b$, its derivative in z is nonzero by hypothesis; the holomorphic implicit-function theorem gives $z = z(u, \lambda)$ and hence $B = B(u, \lambda)$ on a product polydisk. At a negative edge use $t = (C + B - b)/B$ in the same way. Shrinking the polydisk preserves the strict sign of $\Im B$, the nonvanishing physical prefactor and the other strict face conditions. Cauchy estimates on a slightly larger polydisk give the common holomorphic bounds required in Proposition 4.3 and Theorem 5.1. For a negative edge the selected integers may depend on the complex ratio. The equal-coupling boundary is treated by the collinear estimates, and joint holomorphy on interior compact sets also applies in complex-ratio neighborhoods at $\lambda = 1$. □

Proof of Theorem 1.2. The exact open-set equalities are Theorem 2.8. The physical sums of Proposition 3.2 restrict holomorphically to these pullbacks. A pullback boundary point maps to the corresponding joint boundary by the open-set equality and continuity.

Fix $\lambda \neq 1$. Proposition 2.9 gives a dense set on this slice at which the edge map has nonzero derivative and the free phase is a chosen prime-grid value. At each such point the line is a holomorphic curve $B = B(u)$ or $B = B(t)$ of the type used in Theorem 1.1. Apply Proposition 4.3 or Theorem 5.1 to that varying anisotropy. The restricted full sum is nonextendable there. The one-variable collar in Lemma 2.10 and Corollary 2.12 extend the obstruction to every boundary point.

Now let $\lambda = 1$. The exact coefficient and normalization identities in (6.43)–(6.44) identify the restriction with the full even and odd integrals of Section 6; its local obstructions were proved in Proposition 6.10. Lemma 2.13 maps the two slice domains to the exterior and disk, respectively. At every boundary point with $\cosh \zeta \neq 0$ this is a local biholomorphism. The selected obstructing points pull back densely on the nonramified boundary arcs. The remaining points with $\cosh \zeta = 0$ are limits of those arcs. The same local collar therefore proves nonextension everywhere on both slice boundaries.

Section 3 identifies the energy-field response $\chi = \beta \mathcal{X}$, with the plus phase fixed in LT. The complete coefficient sum includes the origin and both axes and subtracts the zero-particle M^2 in LT. The entire factor $\beta = \zeta/(2J_x)$ is nonzero on both boundaries, since zero lies in the HT interior and strictly outside the LT closure. It therefore preserves local boundary extendability. □

7.3 Derivation of the complete isotropic theorems from the slice theorem

Proof of Corollary 1.3. Apply the now proved Theorem 1.2 with $\lambda = 1$. The identities $S = T = s$, $p = s^2$, $B = 1$, and

$$\mathcal{D}(\zeta_1, \eta_1; z, z) = s \left[s + s^{-1} - \tfrac{1}{2}(\zeta_1 + \zeta_1^{-1} + \eta_1 + \eta_1^{-1}) \right]$$

identify the parent coefficients term by term with the full isotropic coefficients. On the real LT interval $\sqrt{p} = s$, so each factor $\sqrt{p}/\mathcal{D}$ is exactly its isotropic counterpart. The even prefactor becomes $(1 - s^{-4})^{1/4}$. On the real HT interval the dispersion is $s + s^{-1} - \cos q$ and the odd prefactor is $(1 - s^4)^{1/4}/s$, giving the exterior auxiliary variable $1/s$. Normal convergence and the identity theorem extend these identities to their physical complex domains.

For HT, the biholomorphism in Lemma 2.13 transports the restricted holomorphic function to the disk. The removable value at zero is one, independently checked at the end of Section 6. For LT, the normalized termwise identities show invariance under every deck translation $z \mapsto z + 2\pi i k$;

the full normalized function therefore descends from $\Omega_{L,1}$ to a single-valued holomorphic function on $|s| > 1$. It is the positive real LT germ and has the stated connected onsite contribution.

Suppose one of these descended functions extended through a point s_0 of the unit circle. Choose a boundary preimage z_0 on its physical slice. Composing the proposed extension with the entire map sinh would extend the slice function near z_0, in agreement on the physical side. This contradicts Theorem 1.2. Composition with sinh also applies at its ramified boundary values $s_0 = \pm\mathrm{i}$. Thus the unit circle is a natural boundary for both normalized germs.

Finally $\beta = z/(2J)$ on the prescribed physical temperature continuation yields the energy-field claims. In the disk it is the branch $\operatorname{arsinh}(s)/(2J)$. In the exterior it is interpreted on the proved covering. At all selected nonaxis points it is analytic and nonzero, so the local singular derivative remains nonzero after the product rule; density gives all remaining boundary points. □

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